From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 1990–2024

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Abstract

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when refining ceased at Matosinhos in May 2021. Using Eurostat annual oil balances for Portugal and Spain, JODI monthly physical flows, DGEG national statistics and the European Commission Weekly Oil Bulletin, this paper measures how the product balance and price exposure changed over 1990–2024.

Portugal's diesel trade position crosses the same line three times: a net exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013 in every year but 2019, and an importer in every year since 2021. The 2013 hydrocracker therefore restored a position the country had held before; it created nothing new, and the 2021 closure returned the system to the dependence it had run in the 2000s. By 2023 domestic manufacturing covers 0.678 of diesel demand and the net-import-to-demand ratio reaches 0.262, which is the dependence Portugal ran in 2009–2011 on half the manufacturing base it had then. Gasoline, which entered each event with output far above domestic demand, moves as far on both ratios but crosses neither threshold. On prices, both Portugal–Spain pairs are cointegrated in log levels and both error-correction speeds increase several-fold after May 2021.

No causal effect is claimed. The closure and the March 2022 European supply shock fall ten months apart, and no design here separates their contributions.

1 Introduction and Research Question

Portugal operated two refineries for most of the study period. Sines, the larger, received a hydrocracker that entered commercial production in January 2013 and was intended to raise diesel yield relative to fuel oil. Matosinhos ceased refining in May 2021 following a December 2020 announcement, concentrating Portuguese refining at Sines. Ten months later, in March 2022, the invasion of Ukraine and the subsequent withdrawal from Russian oil products disturbed European product supply generally, and vacuum gas oil, a diesel feedstock at Sines, in particular.

The empirical question is whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products and reduced supply resilience, and whether Portugal–Spain price co-movement changed after the closure.

The answer this paper arrives at is a sequence of events, no one of which carries it, and the sequence is the argument. Portugal was a net diesel exporter through most of the 1990s, in eight of the nine years from 1990. It crossed into net import at the end of that decade and stayed there for fourteen consecutive years. Refinery output breaks upward around 2000 while neither dependence ratio moves with it, because the system was growing to meet demand that was growing faster. Growth of that kind changes a system's size without changing its character.

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In 2013 the hydrocracker changes the character. It shifts the product slate towards middle distillates and Portugal crosses back into net export, holding that position in every year from 2013 to 2018, with 2019 a single-year interruption and 2020 a brief return. In May 2021 Matosinhos closes and the crossing reverses a third time. Every year since has been a net-import year, and in 2023 the net-import-to-demand ratio reaches 0.262 while domestic manufacturing covers 0.678 of diesel demand. Those are the window's extremes, but narrowly: 2009 reaches 0.2534 and 0.6982 on a system with two refineries. What distinguishes 2023 is not the level but that the level was reached on half the manufacturing base. Ten months after the closure the European energy shock arrives and tests the reduced system, and the test falls almost entirely on diesel.

Read as a sequence, these are not three unrelated breaks but one system crossing and re-crossing a single line. That framing matters for what the 2013 unit did, which was not to move Portugal somewhere new but to restore a trade position the country had held in the early 1990s. And it matters for what followed, because the 2021 closure did not merely undo that restoration. It carried the system past the dependence it had run in the 2000s, to the most import-dependent year in thirty-five. What the paper cannot do is apportion that final movement between the closure and the shock, because they fall ten months apart. That limit is why the monthly design exists and why the causal question is left open.

The paper does not assume that closure raised prices. It separates four mechanisms and tests each in turn:

refining capability → production and trade balance → physical import dependence → price co-movement.

Four hypotheses were pre-specified before estimation:

1. the 2013 hydrocracker changed diesel-producing capability and the trade balance.
2. the 2021 Matosinhos transition changed import dependence and domestic output coverage.
3. 2022 is a distinct supply-system stress episode and cannot serve as clean post- treatment evidence.
4. any price change must be evaluated on pre-tax prices, since tax changes would otherwise be read as pass-through.

No published study of the May 2021 Matosinhos closure was found in preparing this paper, and no peer-reviewed treatment of the 2022 vacuum gas oil disruption as such. The closure is documented here from corporate disclosures and the balance data, and the 2022 chain in Section 8 rests on those disclosures and on the trade record, there being no prior analysis. That absence is the reason for the paper and also a limit on it: there is no independent account against which to check the reading offered here.

A fifth candidate break year, 2000, was added after the annual panel was extended back to 1990 and the 1990s were seen for the first time. It was not pre-specified, and every result concerning it is exploratory. It is reported because omitting a visible feature of the series would be worse than labelling it, but it carries less evidential weight than the pre-specified events and no conclusion here depends on it.

Throughout, four evidence levels are kept distinct: a *documented event*, a *descriptive change*, a *statistical association or structural break*, and a *causal effect*. Only the first three are claimed anywhere in this paper.

2 What Resilience Means Here

In this study, resilience means the capacity of Portugal's domestic refining and its trade channels to absorb a product-specific supply shock without a sharp increase in import reliance or in exposure to prices formed elsewhere. That definition has four moving parts, and each is measurable here: how much of demand domestic manufacturing can cover, how far the trade channel has to stretch when manufacturing falls short, how few places the manufacturing happens in, and how much a system leaning on trade gives up in price independence. A system is more resilient on this reading when a shock of a given size moves those quantities less.

Energy security has no single accepted measure. The reviews that map the field group candidate indicators into availability, affordability, accessibility and acceptability, and note that no study operationalises all of them at once [Kruyt et al., 2009, Ang et al., 2015]. Existing quantitative work concentrates on the dimensions this paper cannot see. Source diversification, political risk in supplier countries and transport risk are the usual objects of measurement [Cohen et al., 2011, Stirling, 1998], and the relationship between import diversification and security is itself contested [Vivoda, 2009]. The definition used here is narrower than any of these and is narrower deliberately: it is the part of the concept a national product balance and a bilateral price relation can actually measure.

Two things follow. The first is that nothing below should be read as a security assessment in the sense those literatures use. The second is that the caveat in Section 13, that rising import dependence is not by itself evidence of reduced security, is not this paper's invention but a standing result in that work [Vivoda, 2009].

The definition is deliberately narrow, and the narrowness is the reason it can be tested. It asks about the observable response of a physical balance and a price relation, not about whether the lights stay on. Four quantities carry it, and the rest of the paper is organised around them.

Two of the four are read off the same annual balance, and both are needed because they ask different questions of it. The domestic supply buffer is refinery output over domestic demand, the share of consumption the country manufactures. It measures capability and not self- sufficiency, since output may be exported while demand is met by imports, which is the normal state for Portuguese gasoline; a ratio above one says the system *could* in principle cover its own demand, not that it does. Import exposure is the other side of the same balance, how much of consumption arrives through the trade channel and in which direction on net. Gross import dependence, imports over demand, gives the volume that must be sourced abroad, while the net-import-to-demand ratio gives the country's position, negative for a net seller and positive for a net buyer. A system can move on one of these and not the other, and Portugal does exactly that around 2000.

Concentration is cruder and harder to argue with: Portugal manufactured in two places until May 2021 and in one afterwards, one of a great many such closures. Of close to a hundred European refineries, twenty-eight above thirty thousand barrels a day were shut or converted after 2009 [FuelsEurope, 2025]. That is a documented event, not an estimated quantity, and Section 2.1 sets out why a shock to a single-site system is not the same object as the same shock to a distributed one.

Price integration is how closely, and how quickly, domestic pre-tax prices track a neighbouring market's. A system meeting more of its demand through imports has less room for domestic price formation, so integration is the price-side counterpart of physical exposure. Section 9.1 onwards measures it as contemporaneous co-movement and as the speed at which a gap against the long-run relation closes.

Several things that belong to a full account of security of supply are absent, and their absence bounds every claim made here. Compulsory stocks and commercial inventory are not observed, and stockholding is the first line of defence against an interruption. Port, storage and pipeline capacity, which determine whether an import volume can physically be landed and moved, are not observed. Nor are supplier diversification, crude slate, contractual terms, or spare capacity at the remaining site. A country that imports a large share of its diesel through deep, liquid, well-connected markets may be more secure than one that manufactures more of its own with a single vulnerable feedstock route. Nothing here can distinguish those cases.

What is measured, then, is the structural dimension: how much of demand the domestic system covers, how much arrives by import, how concentrated the manufacturing is, and how tightly domestic prices move with a neighbour's. Where the paper says dependence rose, it means those quantities moved. It does not mean that supply became less secure. The available data cannot answer that question.

2.1 What the closure changed, beyond capacity

The quantity that changed in May 2021 is capacity, and that is what the rest of this paper measures. But the closure also changed the shape of the system in ways a capacity figure does not express, and those bear directly on how the 2022 shock was met. What follows is structural reasoning about a documented event. Nothing in it is estimated. None of it is measured here, and it is set out because the reader needs it to interpret what is.

With two refineries, an interruption confined to one leaves the other running and the national balance absorbs part of the loss internally. With one, an interruption at that site has no domestic offset at any price, and the entire adjustment must run through trade. Portugal moved from the first situation to the second, and the observed 2022 adjustment did run entirely through trade. The same fact governs planned outages, since refineries require periodic shutdowns and two sites allow these to be staggered so that some domestic conversion capacity is always available; one site means every turnaround removes all of it. No turnaround schedule is observed here, and none of this is specific to Portugal. It matters because an unplanned outage at Sines would leave a signature in annual balances close to the one a feedstock disruption leaves. Maintenance therefore survives as an alternative explanation for 2022 that this paper cannot dismiss.

Flexibility in the product mix went with it. Two plants of different configuration can be run to different slates; one plant's slate is set by its own configuration and the crude available to it, and any further adjustment has to be bought or sold abroad. The Portuguese configurations are not observed here, so how much was lost is unknown. That some was lost follows from there being one plant instead of two.

The 2022 disruption was specific to vacuum gas oil, a feedstock in diesel manufacturing at Sines. Before May 2021 a VGO-specific problem at Sines would have left Matosinhos unaffected. After it, Sines was the whole of Portuguese refining, so a shock to one feedstock at one plant was a shock to national diesel manufacturing. The closure did not merely reduce capacity. It concentrated the capacity and its feedstock vulnerability in the same place, ten months before that vulnerability was tested.

This is why the paper treats the closure and the 2022 shock as parts of one chain instead of rival explanations, and it is why the causal question is left open, for reasons about the system itself.

3 Data

3.1 Sources

Four independent bodies publish the series used here. Their identifiers, roles and retrieval points were frozen at the time of extraction. Full URLs appear in the References.

- **JODI Oil World Database**: monthly secondary-product imports, exports, refinery output and demand [Joint Organisations Data Initiative, 2026]. This is the main physical-flow source because it permits transparent time aggregation and carries per-observation assessment codes.
- **Eurostat `nrg_cb_oil`**: annual oil supply, transformation and consumption for Portugal and Spain [Eurostat, 2026]. This is the only source covering both countries on a common definition.
- **DGEG**, the Portuguese national statistical authority: annual product trade workbooks and a long domestic-sales series [Direção-Geral de Energia e Geologia, 2026b,a]. This is the only source compiled independently of JODI, so it is what makes a genuine cross-check of the trade figures possible.
- **European Commission Weekly Oil Bulletin**: weekly consumer prices with and without taxes from 2005 [European Commission, Directorate-General for Energy, 2026]. Its national methodology note records that the Portuguese figures are volume-weighted averages over roughly three thousand retail stations, reported for the Monday of each week and net of recorded discounts [European Commission, Directorate-General for Energy, 2021]. This is the only price source covering both countries on a common definition and at a frequency that can separate a monthly event from a weekly one.

The four are not alternatives to one another, and none of them could carry the paper alone. Each covers something the others cannot: Eurostat reaches back to 1990 and covers Spain, JODI resolves months but begins in 2002, DGEG is compiled independently of JODI but publishes trade only from 2019, and the Bulletin is the sole source of prices. The arms of the analysis are therefore bounded by different sources, and each runs over a different window as a result.

Galp corporate disclosures supply dated refinery events [Galp Energia, 2013, 2020, 2022]. They are used as event metadata only, never as outcome data, and the announcement date of December 2020 is deliberately not treated as the physical closure date.

3.2 Coverage

Table 1 states what each source covers over the study window.

The annual panel is built from Eurostat, which is the only source reaching back to 1990. JODI begins in 2002 and so cannot carry the annual arm. It supplies the monthly flows and serves as an annual cross-check over the years both cover. The monthly source runs to 2026-06 and is trimmed to the study window, leaving 276 months per product.

3.3 The one material coverage gap

DGEG publishes annual product trade workbooks only from 2019. For 2005–2018 the trade cross- check therefore runs against Eurostat instead of the DGEG primary publication. This is weaker corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both exist. Eurostat is best understood as a second channel for the same national statistics.

Table 1: Source coverage, 1990–2024.

Source	Role	Coverage
Eurostat	Annual balance, PT and ES; the source for the annual panel	Complete for both countries from 1990, 280 of 280 product-flow cells each
JODI	Monthly physical flows; annual cross-check	Complete from 2002; every year carries 12 observed months and assessment code 1
DGEG trade	National trade statistics	**2019–2024 only.** Earlier annual workbooks are not published in this series
DGEG sales	Domestic market sales	Complete, 1970–2024
EC Bulletin	Weekly consumer prices	Complete. 995 Portuguese and 994 Spanish weekly observations per product, 2005-01-03 to 2024-12-30

Domestic demand is unaffected. The DGEG long sales workbook covers the whole window, so demand carries three sources throughout.

3.4 Product definitions

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `O4652`, motor gasoline. Aviation gasoline is excluded from both. In the DGEG sales workbook, gasoline is the *Gasolinas* total, and diesel requires coloured and marked gasoil to be added to road gasoil. Road gasoil alone sits about ten per cent below the JODI and Eurostat demand concept. Comparability is audited against a published product crosswalk, which flags the DGEG trade labels as requiring review before numerical reconciliation.

Differing biofuel treatment does not drive residual differences between sources. The Eurostat blended-biofuel wedge, `O4671` less `O4671XR5220B`, is 0.0 kt on exports in every year of the window and at most 4.5% on imports, so blending cannot explain any export divergence at all.

3.5 Cross-checking the sources against each other

Two of the three trade sources are genuinely independent, and where they disagree the paper reports the result on both. Nine of eighty JODI–Eurostat cells and eight of twenty-four JODI–DGEG cells diverge beyond tolerance. The tolerances, the divergent cells and the reason each is accepted are set out in Appendix B, and no divergent cell enters a headline figure without the alternative value being reported alongside it.

4 Methods

For fuel j in year t, with imports M, exports X, domestic demand D and refinery output Q^{refinery}:

$$\text{NetImports}_{j,t} = M_{j,t} - X_{j,t}, \qquad \text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}},$$

$$\text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}}, \qquad \text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

The refinery-output-to-demand ratio is *not* self-sufficiency. Portuguese refinery output may be exported while domestic demand is met partly by imports, and for gasoline that is the normal state of the system.

Three designs are used on the annual and monthly panels, because they fail in different places and agreement between them is worth more than any of them alone.

Each candidate break is tested inside a window reaching back to the previous candidate, so the 2013 test runs on 2000–2020. That keeps one event out of another's pre-period, at the cost of letting 2000, which Section 6.1 reports as exploratory and not surviving a break-date search, set the sample of a pre-specified test. The 2013 result does not depend on it: the interrupted-trend models in Section 6.2 use the whole panel and find the same break.

Annual breaks are tested by Chow tests on a linear trend at candidate break years, of which 2013 and 2022 were pre-specified and 2000 is exploratory. A Chow test [Chow, 1960] fits two trends and asks whether they beat one. It is therefore demanding of data on both sides of the break, and weak wherever the post-event window is short. 2021 is excluded as a transition year so that it is assigned to neither side, since a year in which the closure happened belongs to neither regime. Twenty-four tests invite false positives, so a Benjamini–Hochberg adjustment [Benjamini and Hochberg, 1995] is applied across all of them. That controls the share of rejections expected to be spurious. It says nothing about the chance of any single one.

The annual interrupted trends are ordinary least squares with a common trend, a post-event level term and a post-event slope term, HAC(1) covariance, 2021 again excluded. This design retains all thirty-five annual observations, less the excluded transition year, instead of splitting the series, which is what lets it say something about 2022 where the Chow test cannot. It buys that power by imposing a single trend across the whole window, so where the two designs disagree the disagreement concerns that assumption.

Monthly event timing comes from a segmented regression [Wagner et al., 2002, Bernal et al., 2017] on 276 monthly observations, 2002–2024, with month fixed effects, a calendar-month trend and HAC(3) errors. Written out, for outcome y_t at elapsed month t:

$$y_t = \beta_0 + \beta_1 t + \sum_{p \in P} \left[\delta_p D_{p,t} + \gamma_p D_{p,t} \left(t - t_p\right)\right] + \sum_{m=2}^{12} \mu_m M_{m,t} + \varepsilon_t, \tag{1}$$

where P is the set of three post-closure phases, $D_{p,t}$ indicates that month t falls in phase p, t_p is that phase's first month, and $M_{m,t}$ are calendar-month indicators. The phases are mutually exclusive, so δ_p is the level shift at the boundary of phase p measured against the extrapolated pre-closure trend $\beta_0 + \beta_1 t$, not against the preceding phase, and γ_p is the slope within phase p. The two must be quoted together, and δ terms from different phases must not be added.

For prices, model family is selected from stationarity and cointegration diagnostics. HAC(8) covariance throughout. Pre-tax prices are the primary measure.

5 Descriptive Evidence

5.1 Summary statistics

Over 1990–2024, Portuguese diesel demand averaged 4,522 kt against average refinery output of 4,348 kt, average imports of 804 kt and average exports of 643 kt. Averaged across the window, then, domestic manufacturing fell short of domestic demand and the country bought more diesel than it sold. Gasoline runs the other way on both counts. The two fuels therefore sit in opposite positions relative to the same demand.

That opposition is worth holding onto, because it governs how each product responds to the same event. A loss of refining capacity acts on a deficit in one case and on a surplus in the other, so an identical proportional loss need not produce a comparable movement in the two balances. Section 5.4 takes up why the slate is built that way, once both panels have been shown.

Peaks are informative about timing, and about which series respond to which event. Diesel refinery output and exports both peak in 2015, two years after the hydrocracker, and diesel imports peak in 2023, two years after the closure. Each production series therefore peaks on the far side of a capacity event, with a lag of about two years in both directions.

Diesel demand peaks in 2004, before either event, and nothing in the capacity record explains that date. The demand path and the capacity path are not the same story, which matters for every ratio reported below, since a ratio moves when either its numerator or its denominator moves and the two are driven by different things.

5.2 Diesel

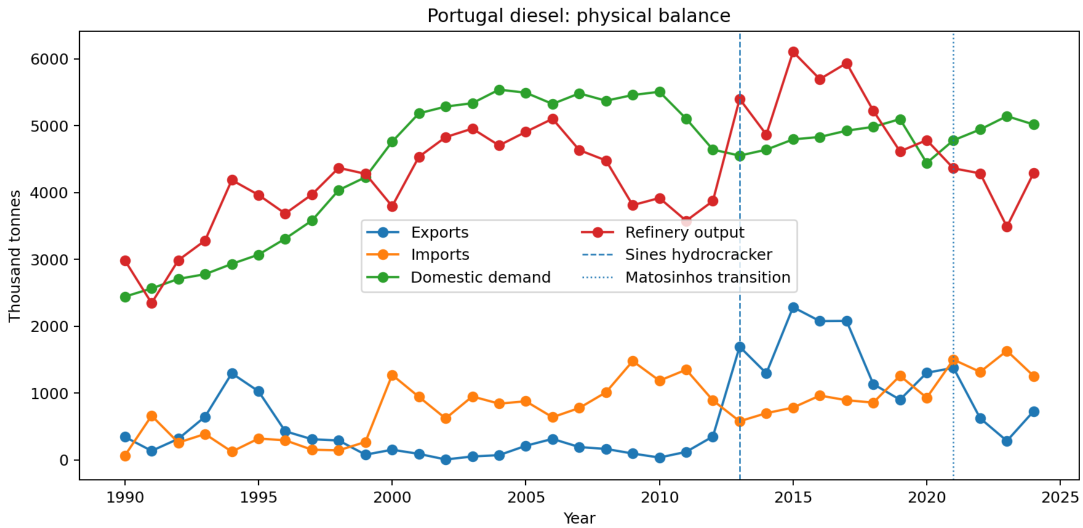

Figure 1: Portuguese diesel imports, exports, refinery output and demand, 1990–2024.

Table 2 gives the annual diesel panel, and the crossings described in Section 1 are visible in the last two columns. Taking the later part of the window, from 2005 to 2012 Portugal is a net diesel importer with output covering 0.70 to 0.96 of demand. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative: Portugal manufactures more diesel than it consumes and exports the surplus. From 2021 the system returns to net import dependence, and by 2023 output covers only 0.678 of demand.

5.3 Gasoline

Gasoline behaves differently (Table 3). Portugal is a net gasoline exporter in every year of the window except 1993, when imports briefly exceeded exports and the net-import-to-demand ratio reached +0.06. Output covers between 0.94 and 2.66 times domestic demand, the low point being that same year. From 1994 onward the ratio never falls below one, and there is no regime change comparable to diesel's. It rises after 2013 and drifts down after 2021 without approaching one again. This is precisely the asymmetry the ratio definition warns about. A system can manufacture a large surplus of one light product while importing another, and reading the ratio as self-sufficiency would obscure that.

Table 2: Portugal diesel annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	64	344	2,442	2,984	0.03	−0.11	1.22
1991	661	135	2,564	2,345	0.26	0.21	0.91
1992	259	315	2,707	2,987	0.10	−0.02	1.10
1993	387	646	2,777	3,281	0.14	−0.09	1.18
1994	123	1,295	2,931	4,186	0.04	−0.40	1.43
1995	319	1,026	3,070	3,963	0.10	−0.23	1.29
1996	293	428	3,309	3,684	0.09	−0.04	1.11
1997	152	309	3,580	3,970	0.04	−0.04	1.11
1998	143	290	4,035	4,368	0.04	−0.04	1.08
1999	268	78	4,228	4,279	0.06	0.04	1.01
2000	1,271	152	4,759	3,794	0.27	0.24	0.80
2001	943	90	5,183	4,533	0.18	0.16	0.87
2002	620	6	5,284	4,827	0.12	0.12	0.91
2003	949	51	5,335	4,955	0.18	0.17	0.93
2004	841	71	5,538	4,703	0.15	0.14	0.85
2005	877	211	5,492	4,906	0.16	0.12	0.89
2006	638	314	5,324	5,102	0.12	0.06	0.96
2007	776	192	5,482	4,634	0.14	0.11	0.85
2008	1,011	164	5,372	4,478	0.19	0.16	0.83
2009	1,478	95	5,457	3,810	0.27	0.25	0.70
2010	1,185	35	5,506	3,917	0.22	0.21	0.71
2011	1,349	120	5,103	3,576	0.26	0.24	0.70
2012	890	350	4,641	3,872	0.19	0.12	0.83
2013	578	1,693	4,551	5,399	0.13	−0.25	1.19
2014	699	1,295	4,640	4,865	0.15	−0.13	1.05
2015	782	2,284	4,794	6,104	0.16	−0.31	1.27
2016	962	2,076	4,830	5,695	0.20	−0.23	1.18
2017	893	2,079	4,924	5,935	0.18	−0.24	1.21
2018	857	1,134	4,981	5,224	0.17	−0.06	1.05
2019	1,258	900	5,097	4,614	0.25	0.07	0.91
2020	929	1,303	4,441	4,783	0.21	−0.08	1.08
2021	1,500	1,377	4,780	4,361	0.31	0.03	0.91
2022	1,316	622	4,942	4,285	0.27	0.14	0.87
2023	1,632	284	5,142	3,487	0.32	0.26	0.68
2024	1,254	728	5,017	4,291	0.25	0.10	0.86

Table 3: Portugal gasoline annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	0	294	1,370	1,704	0.00	−0.21	1.24
1991	122	252	1,507	1,623	0.08	−0.09	1.08
1992	101	244	1,692	1,860	0.06	−0.08	1.10
1993	204	103	1,795	1,687	0.11	0.06	0.94
1994	25	533	1,851	2,354	0.01	−0.27	1.27
1995	34	904	1,890	2,757	0.02	−0.46	1.46
1996	53	658	1,965	2,591	0.03	−0.31	1.32
1997	2	888	1,949	2,835	0.00	−0.45	1.45
1998	0	789	2,027	2,792	0.00	−0.39	1.38
1999	11	588	2,079	2,654	0.01	−0.28	1.28
2000	154	397	2,123	2,296	0.07	−0.11	1.08
2001	10	635	1,995	2,615	0.01	−0.31	1.31
2002	60	540	2,067	2,518	0.03	−0.23	1.22
2003	55	782	2,005	2,732	0.03	−0.36	1.36
2004	130	805	1,927	2,551	0.07	−0.35	1.32
2005	128	792	1,808	2,466	0.07	−0.37	1.36
2006	104	1,270	1,674	2,750	0.06	−0.70	1.64
2007	89	1,091	1,589	2,591	0.06	−0.63	1.63
2008	115	767	1,488	2,091	0.08	−0.44	1.41
2009	240	835	1,462	2,056	0.16	−0.41	1.41
2010	178	1,035	1,388	2,218	0.13	−0.62	1.60
2011	191	799	1,248	1,875	0.15	−0.49	1.50
2012	160	898	1,127	1,864	0.14	−0.65	1.65
2013	102	1,293	1,100	2,338	0.09	−1.08	2.13
2014	154	976	1,092	1,924	0.14	−0.75	1.76
2015	137	1,786	1,080	2,694	0.13	−1.53	2.49
2016	167	1,742	1,069	2,619	0.16	−1.47	2.45
2017	124	1,885	1,032	2,749	0.12	−1.71	2.66
2018	149	1,644	1,027	2,501	0.15	−1.45	2.43
2019	261	1,585	1,067	2,401	0.24	−1.24	2.25
2020	193	1,271	883	1,956	0.22	−1.22	2.22
2021	202	1,424	976	2,146	0.21	−1.25	2.20
2022	230	1,502	1,073	2,247	0.21	−1.18	2.09
2023	347	1,254	1,193	2,063	0.29	−0.76	1.73
2024	300	1,728	1,260	2,660	0.24	−1.13	2.11

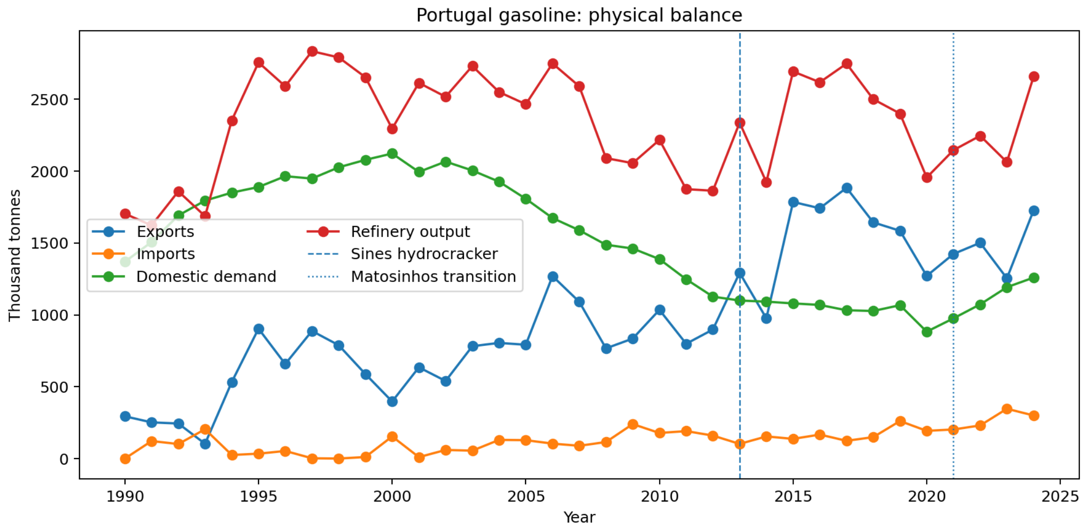

Figure 2: Portuguese gasoline imports, exports, refinery output and demand, 1990–2024.

5.4 Why the two products diverge

The two products behave differently in almost every result that follows, and the reason is visible in the window means. Portuguese diesel demand averaged 4,522 kt against gasoline demand of 1,511 kt, so the country consumes roughly three times as much diesel as gasoline. Manufacturing runs the other way relative to demand. Gasoline output averaged 2,337 kt against that 1,511 kt of demand, a surplus exported at an average of 971 kt a year against imports of only 129.5 kt. Diesel output averaged 4,348 kt against 4,522 kt of demand, a deficit covered by imports averaging 804 kt against exports of 643 kt.

This is the familiar European mismatch in national form. A refinery converts crude into a slate of products in proportions set by the crude and the conversion units installed, and that slate is not free to follow demand. Portuguese demand dieselised, as it did across Europe and for the same fiscal reason: road diesel was taxed below gasoline in almost every OECD country over this period [Harding, 2014], and Portugal is no exception [Silva et al., 2022]. Gasoline demand peaks in 2000 and no later year matches it, while diesel demand peaks in 2004. The gap between the two widened over the window instead of holding fixed: diesel ran at under twice gasoline demand in the early 1990s and at around four times by its end. The refining system was left producing more gasoline than the country could burn and less diesel than it needed, so gasoline became a structural export and diesel a structural import. This is the European pattern, not a Portuguese peculiarity: the continent's refineries over-produce gasoline and under-produce diesel because national fiscal policy favoured diesel, and a system with few sites is hit harder when one of them closes [de Boncourt, 2013].

That difference in starting position, and not any difference in how the two products were treated, is what makes them respond differently to the same events. A hydrocracker converts heavy gasoil into middle distillates, so the 2013 unit acted on the deficit product: diesel output peaks at 6,104 kt in 2015 and the trade position crosses into surplus. Gasoline had no deficit to close. Its ratios do break at 2013, and by more than diesel's: the net-import-to- demand level change is −0.713 against diesel's −0.540. What gasoline does not do is cross anything, because it was already a large net exporter and stayed one.

The same asymmetry governs the losses. Gasoline entered the 2021 closure with output well above domestic demand, so removing capacity moved the ratio down a wide margin without reaching one, and the gasoline refinery-output-to-demand ratio stays above 1.7 in every year after the closure. Diesel entered with a ratio near one and no comparable margin, so the same proportional loss pushed it to 0.678 by 2023 and turned the country back into a net buyer. A buffer is only protective while it is large relative to the shock, and Portugal held a large one in the product it did not much need and a thin one in the product it did.

One question the ratios do not answer on their own is why Portugal keeps manufacturing gasoline it does not consume instead of making less of it. The answer is that it largely cannot. A refinery's products come out of the same barrel in proportions bounded by its configuration, so gasoline output is not an independent decision: cutting it means cutting throughput, and cutting throughput means making less diesel too. A country running its plants at the rate its diesel demand requires will produce more gasoline than it needs, and the surplus is exported because the alternative is not producing the diesel. That inverts the natural reading of the gasoline figures: the persistent export surplus is not a sign of strength in gasoline but a residue of running the system for diesel. It also explains why gasoline coverage held up through the closure while diesel did not. Gasoline output tracks throughput, Sines continued to run, and gasoline demand is small enough relative to that throughput that a large proportional loss of national capacity still left output well above it. Diesel had no such cushion because diesel demand is what sets the throughput in the first place.

This account is consistent with every result reported below, but it is not tested by any of them. Yield economics, the crude slate and the commercial logic of export contracts are not observed here, and a competing account in which the two products differ for reasons of contracting and not of chemistry would fit the same series.

5.5 Dependence ratios

Figures 3 and 4 put the three ratios of Section 2 on the same axis for each product, and the contrast between them is the physical result this paper rests on. Diesel coverage crosses one repeatedly: it sits above one in sixteen of thirty-five years, rises through it in 2013 and falls back below it after 2021, reaching 0.678 in 2023. Gross import dependence moves with it, from 0.026 in 1990 to 0.317 in 2023. Gasoline coverage crosses one once in thirty-five years, dipping to 0.94 in 1993, and otherwise runs between there and 2.663. The two products are therefore not two instances of the same series with different noise. One spends the window near a threshold and the other spends it far above one, and every asymmetry reported below follows from that.

5.6 Event years

Table 4 extracts the years that bracket each documented event, together with the refining regime in force. The 2021 row is a transition year and is excluded from the annual before/after designs.

6 Structural Breaks

6.1 Chow tests

Twenty-four Chow tests are run across three candidate break years, and Table 5 lists the six that survive Benjamini–Hochberg adjustment. They fall into two groups, and only the first was pre-specified.

Three concern 2013 and move in the direction the engineering implies: more diesel manufactured, more exported, and a net-import position that turns negative. Extending the series to 1990 strengthens them, because the pre-event segment now carries thirteen observations, up from eight.

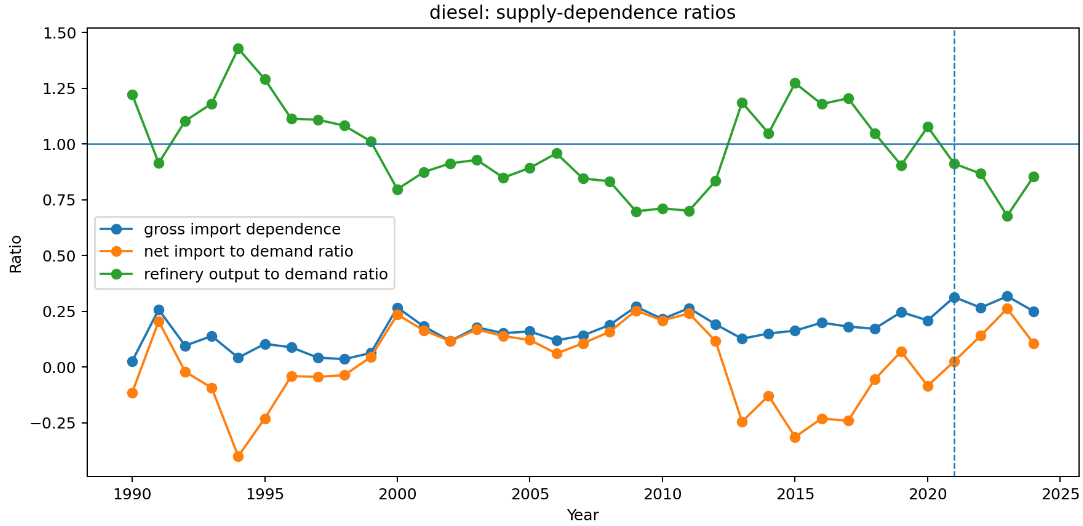

Figure 3: Diesel gross import dependence, net imports to demand, and refinery output to demand. The output ratio crossing one in 2013 and falling back after 2021 is the central physical result of this paper.

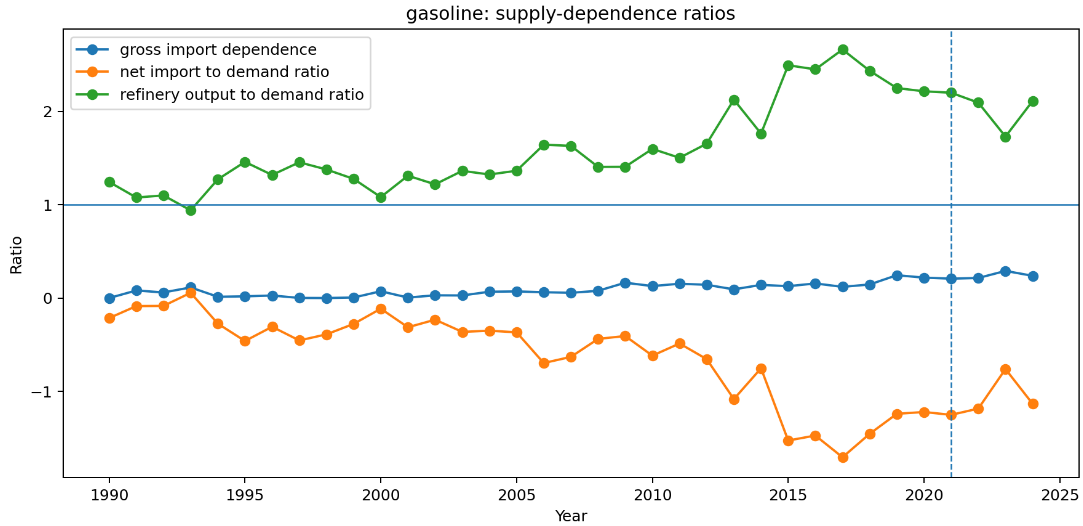

Figure 4: Gasoline dependence ratios. The output ratio dips just below one in 1993 and stays well above it in every other year, so no regime change comparable to diesel occurs.

Table 4: Balance at the event years. The two independent trade sources disagree on the 2022 diesel export cell. The table carries the national figure of 622.2 kt, which puts net imports over demand at +0.140; the monthly source reports 331.0 kt, which would put it at +0.199. See Table 16.

Year	Product	Exports	Imports	Demand	Ref. out.	Net imp./dem.	Regime
2012	diesel	350	890	4,641	3,872	+0.116	two refineries
2013	diesel	1,693	578	4,551	5,399	−0.245	two refineries
2020	diesel	1,303	929	4,441	4,783	−0.084	two refineries
2021	diesel	1,377	1,500	4,780	4,361	+0.026	transition
2022	diesel	622	1,316	4,942	4,285	+0.140	Sines only
2023	diesel	284	1,632	5,142	3,487	+0.262	Sines only
2024	diesel	728	1,254	5,017	4,291	+0.105	Sines only
2012	gasoline	898	160	1,127	1,864	−0.655	two refineries
2013	gasoline	1,293	102	1,100	2,338	−1.083	two refineries
2020	gasoline	1,271	193	883	1,956	−1.221	two refineries
2021	gasoline	1,424	202	976	2,146	−1.252	transition
2022	gasoline	1,502	230	1,073	2,247	−1.185	Sines only
2023	gasoline	1,254	347	1,193	2,063	−0.761	Sines only
2024	gasoline	1,728	300	1,260	2,660	−1.134	Sines only

The other three concern 2000, which is exploratory, having been added after the extended panel revealed the 1990s. Refinery output breaks upward for both products, most strongly for gasoline, and diesel imports break upward with it. What does not break at 2000 is either ratio: neither the net-import-to-demand nor the refinery-output-to-demand ratio shows a discontinuity there. Output and imports both rose because demand rose faster, so the system was expanding without changing character. This is a different kind of event from 2013, and treating the two alike would misread it.

The 2000 results need a different test from the one reported for them. A Chow test is valid at a break specified in advance, and 2000 was chosen after the extended panel was seen. Searching the 1990–2012 window over candidate break years and comparing the largest statistic with its own simulated null, in the manner of Andrews [1993], puts diesel imports at $p = 0.088$, so the break the paper reports at 2000 does not survive being treated as what it is. The two refinery-output series do reject, decisively, but the search does not put their break at 2000: it selects 2007 for diesel and 1995 for gasoline. What the exploratory evidence supports is that something happened in the 1990s and early 2000s, not that 2000 is when it happened. Nothing in the paper's conclusions rests on it. A joint multiple-break estimator [Bai and Perron, 1998, 2003] locates all the breaks at once.

Table 5: Chow tests at candidate break years, 2021 excluded as a transition year. 2013 and 2022 were pre-specified; 2000 was added after the panel was extended back to 1990 and is exploratory.

Product	Outcome	Break	F	p	BH p	$n_{\text{pre}}/n_{\text{post}}$
gasoline	refinery output	2000	22.00	< 0.001	< 0.001	10/13
diesel	exports	2013	21.58	< 0.001	< 0.001	13/8
diesel	net imports / demand	2013	21.40	< 0.001	< 0.001	13/8
diesel	refinery output	2000	11.68	< 0.001	0.0030	10/13
diesel	refinery output	2013	8.23	0.0032	0.0152	13/8
diesel	imports	2000	5.81	0.0108	0.0431	10/13

A joint estimator answers a different question from the one the Chow tests ask, and it is worth asking. Testing candidate years separately asks whether a given year is a break on the assumption that no other break exists, which is false for any series with more than one. Minimising the residual sum of squares over every admissible combination of up to three breaks and selecting the number by BIC [Bai and Perron, 1998, 2003] gives a different kind of answer: across the eight product-outcome series, 2013 is selected in four, more than any other year. The 2000 candidate is selected in one. Whatever happened around 2000 is not what a joint search finds when it is free to place breaks anywhere.

Two limits on that comparison. The search trims five observations from each end and between breaks, so the candidate years run 1995 to 2019 and 2022 lies outside what it can see; nothing in it bears on the closure or the stress episode. And BIC selects three breaks for five of the eight series, several at years this paper does not discuss, which is a reminder that these are short annual series with a good deal of movement in them and that a break-selection procedure will find structure in most of it.

6.2 Interrupted trends, and why they disagree about 2022

The absence of a detected 2022 Chow break is specific to that test. A Chow test asks whether two independently fitted trends beat one. With three post-event annual observations it cannot answer that question. An interrupted-trend model asks whether the level shifts against a trend estimated from the whole series, a far weaker demand on the post-event data. On that specification, reported in Table 6, 2022 is detectable for diesel but not for gasoline.

The window matters here more than anywhere else in this paper. Fitted from 2005 the 2022 diesel export shift is roughly twice as large as it is when fitted from 1990 against a trend informed by the 1990s, where it is −782 kt, and the gasoline shifts lose significance altogether. The longer series is the more honest one, and it makes the annual case for 2022 weaker than a short window implies. The monthly design of Section 7 remains the stronger instrument for that period.

Table 6: Interrupted-trend models, annual series, HAC(1), $n = 34$, 2021 excluded. Intervals are ninety-five per cent. The 2022 rows are reported twice: once as a single-event model and once holding the 2013 break constant. The 2022 export rows also depend on trade cells the two sources dispute; see Table 16.

		Level change			Slope change
Product	Outcome, event	Estimate	Interval	p	Estimate (p)
diesel	exports, 2013	+2,005.5	[1,417.8, 2,593.3]	< 0.001	−117.1 (0.001)
diesel	net imp./dem., 2013	−0.540	[−0.671, −0.408]	< 0.001	+0.029 (0.001)
gasoline	exports, 2013	+411.1	[21.6, 800.7]	0.039	−22.4 (0.355)
gasoline	net imp./dem., 2013	−0.713	[−1.133, −0.294]	< 0.001	+0.038 (0.157)
2022, single-event model					
diesel	exports, 2022	−781.7	[−1,453.0, −110.4]	0.022	+15.0 (0.872)
diesel	net imp./dem., 2022	+0.201	[0.019, 0.383]	0.031	−0.017 (0.616)
gasoline	exports, 2022	−270.9	[−612.9, 71.1]	0.121	+69.3 (0.413)
gasoline	net imp./dem., 2022	+0.307	[−0.053, 0.667]	0.095	+0.070 (0.457)
2022, holding the 2013 break constant					
diesel	exports, 2022	−570.0	[−1,203.5, 63.6]	0.078	+150.1 (0.166)
diesel	net imp./dem., 2022	+0.152	[0.008, 0.295]	0.038	−0.052 (0.149)
gasoline	exports, 2022	−312.2	[−923.0, 298.6]	0.316	+82.1 (0.339)
gasoline	net imp./dem., 2022	+0.480	[−0.068, 1.027]	0.086	+0.066 (0.486)

The 2022 rows appear twice for a reason that changes one of them. A single-event model fits one trend across the whole panel and measures 2022 against it, and for diesel exports that trend is drawn through the 2013 hydrocracker, which is the largest feature of the series. Holding the 2013 break constant moves the 2022 diesel export shift from -781.7 to -570.0 and its p-value from 0.022 to 0.078, so the shift no longer clears five per cent and its interval now includes zero. The misspecification is visible in the residuals: the single-event model has an adjusted R^2 of 0.178 and a Durbin–Watson statistic of 0.527, against 0.731 and 1.271 once 2013 is held constant. The diesel net-import ratio survives the same control, at $+0.152$ with $p = 0.038$. The annual evidence for a 2022 diesel export break is therefore weaker than the first specification suggests. That agrees with the Chow tests and with the monthly design carrying that period.

The 2013 rows repay reading with their slope terms alongside their level shifts. Diesel exports step up by 2,005.5 kt and then decay at 117.1 kt a year, so the fitted effect is about 1,186 kt by 2020 and continues to fall. The hydrocracker did not move the series to a new level and hold it there; it produced a surplus that eroded as demand grew back into the new capacity. By 2019 the erosion had gone far enough to return a net-import year, before the closure had happened at all. The decay term is well identified, at $[-185.6, -48.6]$.

The 2022 gasoline export row does not reach five per cent under either trade source once the panel is extended, as Section B.3 sets out. It is not relied on in the conclusions.

The two designs are not in conflict about the world. They are in conflict about what three annual observations can support. Any statement that annual evidence detects no 2022 break should therefore be read as specific to the Chow specification. The interrupted-trend models and the monthly design of Section 7 agree with each other in sign and rough magnitude.

7 Monthly Event Analysis

7.1 Phase means

The monthly panel is partitioned into four phases: before May 2021, the Matosinhos transition from May 2021 to February 2022, the energy-stress period from March 2022 to December 2022, and post-stress from January 2023. Table 7 gives diesel phase means.

Table 7: Diesel monthly means by event phase, kt per month unless noted.

Phase	Months	Imports	Exports	Ref. out.	Net imp./dem.
Pre-closure	232	75.4	65.4	411.5	0.013
Matosinhos transition	10	133.3	63.0	321.9	0.162
Energy stress 2022	10	111.1	27.7	324.6	0.193
Post-stress	24	117.5	42.3	311.7	0.181

Mean monthly diesel imports rise by roughly a half to three quarters between the pre-closure phase and every subsequent phase, refinery output falls by a fifth and then a quarter, and the net-import-to-demand ratio moves from approximately zero to 0.18.

Two features of that pattern carry more than the individual figures. All three terms move in the directions a capacity loss would produce, with imports up, output down and the balance swinging from roughly even to buying about a fifth of consumption abroad. And no later phase recovers the pre-closure position, so the means describe a level change; the system did not work this one through. They remain raw averages that take no account of seasonality or trend, and a seasonal pattern alone could account for part of a phase difference this size. The model below controls for both.

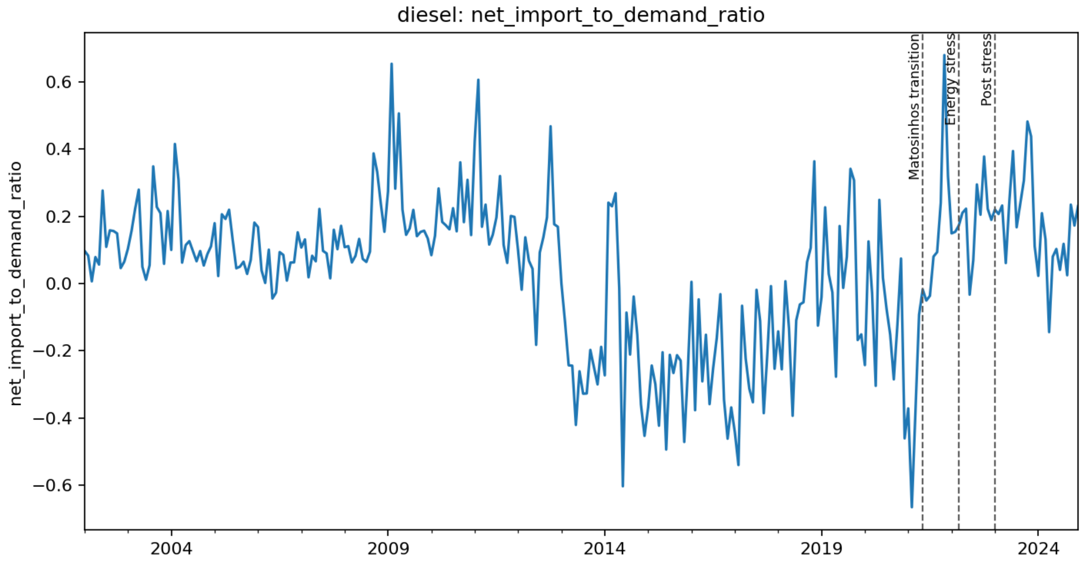

Figure 5: Monthly diesel net imports to demand, with the May 2021, March 2022 and January 2023 phase boundaries marked.

Table 8: Segmented monthly event model, diesel, $n = 276$ months (2002–2024), month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary against the extrapolated pre-closure trend.

		No 2013 term		2013 held constant	
Outcome	Term	Estimate	p	Estimate	Interval
Refinery output (kt/month)	2013 control	—	—	+167.92	[121.3, 214.5]
	Matosinhos transition	−102.89	0.005	−70.52	[−150.4, 9.4]
	phase trend	−4.04	0.553	−2.88	[−16.5, 10.7]
	Energy stress 2022	−148.25	< 0.001	−101.10	[−156.4, −45.8]
	phase trend	+5.16	0.142	+5.50	[−0.7, 11.7]
	Post-stress	−168.45	< 0.001	−110.52	[−177.6, −43.4]
	phase trend	+2.72	0.022	+3.67	[1.2, 6.2]
Net imports / demand	2013 control	—	—	−0.4379	[−0.561, −0.315]
	Matosinhos transition	+0.1923	0.007	+0.0878	[−0.081, 0.256]
	phase trend	+0.0382	0.012	+0.0349	[0.005, 0.065]
	Energy stress 2022	+0.3658	< 0.001	+0.2198	[0.065, 0.374]
	phase trend	+0.0098	0.171	+0.0085	[−0.004, 0.021]
	Post-stress	+0.4951	< 0.001	+0.3174	[0.121, 0.514]
	phase trend	−0.0057	0.097	−0.0086	[−0.016, −0.001]
Adjusted R^2, refinery output		0.2163		0.4046	
Adjusted R^2, net imports / demand		0.2931		0.4982	

7.2 Segmented model

On the specification that holds 2013 constant, the monthly design separates the phases unevenly. The energy-stress period sits −101.10 kt per month below the extrapolated pre-closure trend in refinery output and +0.2198 above it on the net-import-to-demand ratio, both firmly estimated. The post-stress period sits −110.52 and +0.3174, also firmly estimated. The Matosinhos transition does not clear five per cent on either outcome, at −70.52 ($p = 0.084$) and +0.0878 ($p = 0.307$), and its intervals contain zero. So the monthly design does what annual data cannot, in separating a May 2021 event from a March 2022 one, but what it resolves clearly is the stress period and the level that followed, and not the ten months between the closure and the shock. The import level is weaker still, and its energy-stress term reaches five per cent ($p = 0.015$) while its Matosinhos term reaches only ten ($p = 0.081$), so the import channel is not relied on for the closure itself.

Reading the same model without the 2013 term gives larger numbers on every phase, and those are the ones the paper reported until they were checked. The transition reads −102.89 ($p = 0.005$) and +0.1923 ($p = 0.007$), the energy stress −148.25 and +0.3658. The next paragraph is about why the difference arises and which set to believe.

Diesel refinery output sits about 103 kt per month below the extrapolated pre-closure trend during the transition and about 148 kt per month below it during the stress period. The second figure is measured against the same counterfactual as the first, not against the transition phase, so the two do not add. The net-import-to-demand ratio sits 0.19 above that counterfactual at the closure and 0.37 above it during the stress phase, and the closure phase also carries an upward slope of 0.038 per month, so on this fit dependence continued to widen within that phase instead of stepping once and holding.

That slope has a consequence the level shifts alone do not show, and it is worth stating because it looks wrong at first sight. Carried to the last month of the transition, the fitted deviation is $0.1923 + 9 \times 0.0382 = 0.536$ above the pre-closure counterfactual. The energy-stress phase then begins at 0.366 above the same counterfactual. The fitted path therefore steps *down* by about 0.17 in the month the shock arrives. This is a property of the parameterisation in Equation 1, not a claim that dependence eased in March 2022: each δ_p is measured at its own boundary against the pre-closure trend, and a phase entered with a steep slope can end above the level at which the next phase starts. It is also a warning about the slope itself, which is fitted on the ten months of the transition and carries a standard error of 0.0152 against an estimate of 0.0382. Extrapolating it nine months is at the edge of what ten observations support, and no claim in this paper rests on the extrapolated value.

The monthly panel opens in 2002 and the 2013 hydrocracker sits inside it, so the pre-closure trend every phase coefficient is measured against was being fitted through a step of +167.92 kt per month in refinery output. The annual models had the same problem, diagnosed in Section 6.2, in a design that is more exposed to it, and the fit says so: adjusted R^2 rises from 0.2163 to 0.4046 for refinery output and from 0.2931 to 0.4982 for the ratio once the step is held constant.

Holding it constant does not overturn the result but it does redistribute it. The energy-stress shift in refinery output falls from −148.25 to −101.10 kt per month and stays firmly estimated, and the ratio shift falls from +0.3658 to +0.2198. What does not survive is the Matosinhos transition term, which goes from −102.89 ($p = 0.005$) to −70.52 ($p = 0.084$) on output and from +0.1923 ($p = 0.007$) to +0.0878 ($p = 0.307$) on the ratio. On the corrected specification the monthly design detects the 2022 stress and the post-stress level clearly and the ten-month transition only weakly. This reading is narrower than the paper previously offered, and it is the one the corrected model supports.

The fit statistics belong here too. Holding 2013 constant, the adjusted R^2 is 0.4982 for the net-import-to-demand ratio and 0.4046 for refinery output, against 0.2931 and 0.2163 without it. The corrected model explains about half the variation in these series and the uncorrected one about a quarter, and even the corrected figure leaves a great deal of monthly movement the phase structure does not account for.

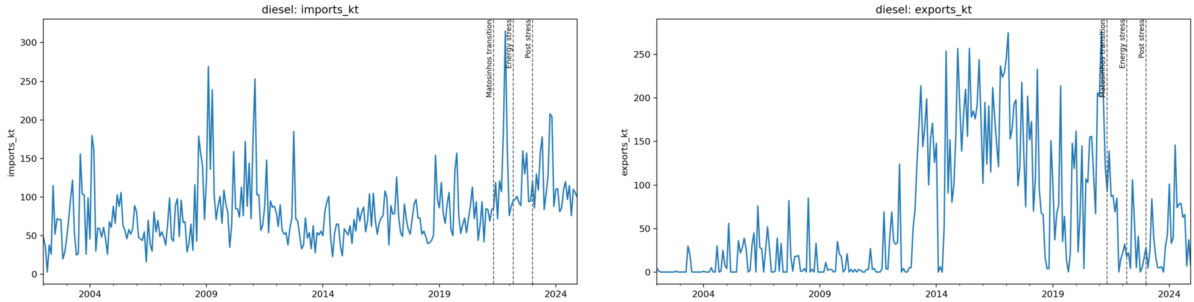

Figure 6: Monthly diesel imports (left) and exports (right).

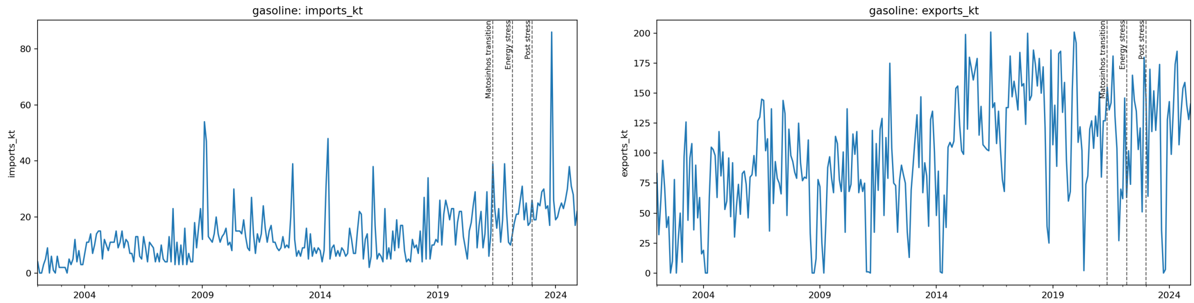

Figure 7: Monthly gasoline imports (left) and exports (right).

8 The 2022 Stress Episode

The chain this section tests runs as follows. Russia supplied a large share of the world's vacuum gas oil. VGO is a feedstock in diesel manufacturing at Sines. Galp announced the suspension of Russian oil-product purchases in March 2022. If that disruption bound, diesel manufacturing should fall while gasoline manufacturing does not, because the constraint is on a diesel feedstock and not on running the plant. Domestic demand still has to be met, so exports should give way first and imports rise second. And because Portugal had been a one-refinery system for ten months, none of that adjustment can be taken up domestically. Each link is testable against the balance, and each holds.

Against a 2013–2019 baseline, 2022 diesel exports sit at the zeroth percentile of the baseline distribution and diesel imports of 1,316 kt sit 2.10 standard deviations above the mean, at the hundredth percentile. Neither number should be read as it stands, and the next two paragraphs say why before the argument continues.

Table 9: 2022 diesel exports against the 2013–2019 baseline, computed on each source.

Source for the 2022 cell	Value (kt)	z	Baseline percentile
Monthly source, flagged as the outlier	331.0	−2.44	0
National statistics, corroborated	622.2	−1.89	0

Neither the level scores nor their detrended counterparts should be leaned on, and the reason is

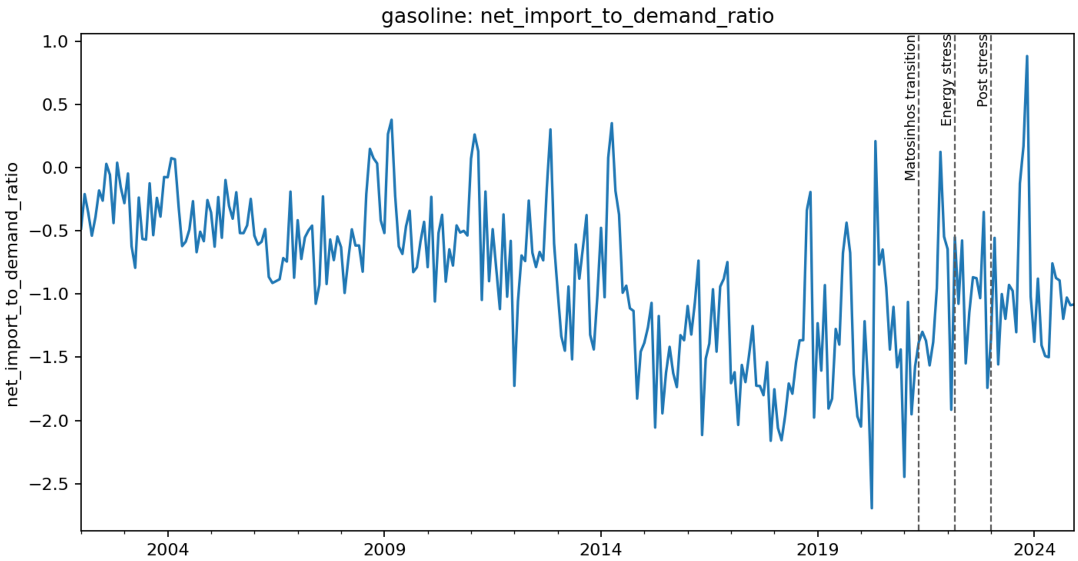

Figure 8: Monthly gasoline net imports to demand. The series sits below zero in nearly every month, with occasional single months where imports exceed exports, consistent with the annual picture of a persistent export surplus.

the same for both. A z-score against a level mean measures the baseline trend whenever the baseline has one, and these baselines trend: diesel imports rise across 2013–2019 at 88.05 kt a year at $t = 4.11$, and diesel demand rises at $t = 16.83$, the steepest of the five. But correcting for that requires extrapolating a 2013–2019 trend across 2020 and 2021, which is not a credible operation for any series here. Done for demand it puts 2022 roughly fifteen residual standard deviations below expectation, which is a reductio on the procedure. The detrended import score of -0.65 is therefore not a more honest version of the level score of $+2.10$; the two are wrong in different directions and the honest statement is that a seven-year baseline interrupted by a pandemic cannot place 2022 at all.

The prediction interval says the same thing in a way that does not depend on the trend. Seven years support an interval for diesel imports running from 295 to 1,427 kt, and 1,316 falls inside it, as do exports, net imports and refinery output on each of the three baseline lengths. A percentile of zero or of one hundred out of seven observations is a statement about seven numbers.

So this section does not carry the 2022 result and should not be read as though it did. The instrument for 2022 is the monthly segmented model of Section 7, which fits a trend and measures each phase against its extrapolation. What remains here is a description, and the description is the argument: the four terms of the balance move together in the particular pattern a feedstock constraint implies.

Gasoline exports in the same year are unremarkable, at $z = -0.18$ and the 29th percentile, and gasoline refinery output is only mildly low ($z = -0.76$). Gasoline imports are elevated ($z = 1.46$), but the gasoline net-import position is not ($z = 0.41$). Diesel demand in 2022 is close to baseline ($z = 0.58$), so the diesel import surge is not demand-driven. Portuguese diesel demand is in any case found to be inelastic to price and driven by economic activity, given how much of it is commercial [Silva et al., 2022, 2021], so a price-induced demand response is not the obvious competing explanation either.

The stress is therefore diesel-specific on the supply side. That is consistent with the documented suspension of Russian oil-product and vacuum gas oil purchases feeding diesel manufacturing at Sines, and inconsistent with a general refining shutdown, which would have moved gasoline equally. The metrics are reported across three baseline windows and the ranking is unchanged in each.

The pattern has to be read as a whole, because what identifies where the constraint bound is the way the four terms of the balance move together. Statistic by statistic it says much less. Refinery output falls to the bottom of its baseline range while gasoline output, low but well inside its own range, does not. That locates the problem in diesel manufacturing while the refinery kept running. Demand sits at its baseline, so nothing was rationed and no adjustment came from consumption. Exports collapse and imports reach the top of the baseline range. A system that cannot make as much of a product, and whose consumers want the usual amount of it, has two ways to close the gap: stop selling it abroad and start buying it. Portugal did both, in that order of magnitude.

This is what the trade channel absorbing a shock looks like, and it is the first direct observation in this paper of resilience in the sense of Section 2 being tested. The domestic buffer contributed nothing, because output moved the wrong way, as a feedstock disruption makes it do. The whole adjustment ran through imports. Whether that counts as a system coping or a system exposed depends on a judgement this paper cannot make, because the terms on which those imports were obtained, and whether they would have been available in a tighter market, are not in the data.

The export figure needs care, because 2022 diesel exports is the single largest disagreement between the two independent trade sources, and the reconciliation identifies the monthly source as the outlier for that cell. Table 9 therefore reports the statistic on both values. The qualitative result is unchanged, since every baseline year exceeds either figure, but the magnitude is not. The standardised deviation is -2.44 on the monthly source and -1.89 on the corroborated national figure. The weaker of the two is the one this paper relies on.

The concentration point matters here in a way it does not elsewhere. A disruption to one feedstock route is a different object in a two-site system than in a one-site system: with two conversion sites, a problem specific to one of them can be partly offset by the other, and with one there is no domestic offset available at any price. Portugal entered 2022 having become a one-site system ten months earlier. The evidence here cannot separate how much of the 2022 movement is the shock and how much is the reduced system meeting it, and that is not a technicality, since the two are not independent. A closure changes how a subsequent common shock propagates, so the common-shock account and the closure account are not competing explanations to be weighed against each other but two parts of one causal chain that this design cannot decompose.

9 Prices

9.1 Diagnostics and model choice

Weekly pre-tax prices are the primary measure. The retail fuel price literature is largely concerned with asymmetry in how quickly prices rise and fall [Bacon, 1991, Borenstein et al., 1997, Peltzman, 2000], a finding whose robustness has itself been questioned [Bachmeier and Griffin, 2003] and whose methods have been surveyed with the warning that the field has been driven by technique and not by theory [Meyer and von Cramon-Taubadel, 2004]. No asymmetry is estimated here. The closest published relative to what follows runs stationarity, cointegration and stability diagnostics on euro-area consumer fuel prices from this same bulletin before choosing a specification [Meyler, 2009], and the practice of testing a long-run coefficient against a value implied in advance rather than against zero follows Bachmeier and Griffin [2006]. One difference from Meyler [2009] is better stated here than left for a reader to find. He models these prices in raw levels and reports that the choice of scale mattered to his results; the diagnostics in

Section 9.1 lead the other way here, and Section 9.2 sets out the evidence for preferring logs, including a sign reversal on the levels specification that the log models do not produce. The disagreement is about which scale the long-run relation lives on. The procedure is the same in both papers. The bulletin does not publish in every week: of the 1,044 weeks between the first and last observation, 995 are present for Portugal and 994 for Spain, and the gaps appear as 25 intervals of a fortnight and 12 of three weeks. Missing weeks are dropped, not interpolated, so a differenced observation spanning a gap is a two- or three-week change entered as one. The unit-root and HAC lag structures both assume even spacing and neither is adjusted for this. The limitation belongs to the price arm, not to the estimator. The diagnostics are run on log levels, which is the scale the models are estimated on, since the short-run specification regresses $\Delta \log \text{PT}$ on $\Delta \log \text{ES}$, and the error-correction model takes its long-run residual from log PT on log ES. An earlier version of this analysis ran them on the EUR per 1000 litres levels, and the two scales do not agree where it matters: diesel Engle–Granger is 0.057 on the EUR levels and below 0.001 on logs, which is the difference between fitting an error-correction model and declaring the pair not cointegrated. The log-scale result is the one reported here.

Each test licenses a different model, which is why they run in this order. A series with a unit root wanders without returning to a fixed level, and regressing one such series on another produces a strong-looking relationship whether or not any relationship exists [Granger and Newbold, 1974]. So the first question is whether the price levels are stationary, because a plain regression is safe only if they are. If they are not, the second question is whether the two wander together, which is what cointegration means and what makes an error- correction model the right form.

Augmented Dickey–Fuller tests [Dickey and Fuller, 1979, Said and Dickey, 1984] on the log levels do not reject a unit root for diesel in either country, at $p = 0.123$ for Portugal and $p = 0.119$ for Spain. For gasoline they reject marginally, at $p = 0.019$ and $p = 0.028$, and that verdict is not quite robust: of the six combinations of deterministic term and lag rule tried per series, the Spanish one fails to reject in the case with a trend and lags by BIC, at $p = 0.074$. Across all of them, no series rejects at one per cent, the bar that matters here.

A marginal rejection is weak evidence in a test whose null is the unit root, so the levels are also tested with that null reversed. KPSS [Kwiatkowski et al., 1992] rejects stationarity for all four series, under both a constant and a constant with trend, at the one per cent level. The two tests therefore agree that these are integrated series, and the marginal gasoline verdict does not survive the reversal.

A levels regression is admitted only on rejection at one per cent rather than five, because the two errors are not symmetric. Regressing one near-integrated series on another is the spurious regression case and its inference is invalid, whereas an error-correction model whose series turn out to be stationary retains a valid stationary regressor and nests the difference specification. Neither product clears that bar.

Both pairs are then cointegrated in logs: diesel Engle–Granger $p < 0.001$ and gasoline $p = 0.0026$. An error-correction model is therefore indicated for both, and estimated in Section 9.4.

9.2 The levels model, and why it is not used

The levels regression was estimated and is retained in the results archive, flagged as not valid for inference for both products. It is reported here because discarding it silently would hide how much specification choice matters, not because it supports any claim.

On that specification the diesel interaction term ES $\times$ post is -0.082 ($p < 0.001$): Portuguese prices appear to become *less* responsive to Spanish prices after the transition. The licensed error-correction model gives $+0.662$ ($p < 0.001$) on the same interaction, the opposite sign. The levels model also reports a deterministic trend of -0.015 per week ($p = 0.005$). Both features are what a regression between two integrated series estimated without its cointegrating relationship is expected to produce. No conclusion in this paper rests on it, and the sign reversal is the reason the stationarity bar in Section 9.1 is set where it is.

9.3 Short-run co-movement

Table 10: Contemporaneous elasticity of Portuguese to Spanish pre-tax prices, $\Delta\log$, HAC(8), $n = 993$ weeks.

Product	Term	Estimate	Std. error	p
diesel	$\Delta\log$ ES	0.7348	0.0341	< 0.001
	$\Delta\log$ ES $\times$ post	0.2638	0.0632	< 0.001
	post	-0.0001	0.0010	0.900
gasoline	$\Delta\log$ ES	0.7997	0.0593	< 0.001
	$\Delta\log$ ES $\times$ post	0.0066	0.1897	0.972
	post	-0.0001	0.0013	0.956

This is the difference-only specification. It is nested in the error-correction models below, which are the ones the diagnostics licence, and it is reported because it is what a reader would obtain without the long-run term and because the two give different short-run elasticities. On this specification the weekly elasticity of Portuguese to Spanish pre-tax diesel prices rises from 0.73 to 1.00. The gasoline interaction is small and indistinguishable from no change. Omitting the disequilibrium term is not innocuous: for diesel it moves the post-transition elasticity from 1.38 to 1.00, because part of the adjustment it leaves out is loaded onto the contemporaneous coefficient.

9.4 Error correction

Both pairs are cointegrated in logs, so for both the short-run coefficient is not the whole result. Table 11 reports the two-step Engle–Granger models [Engle and Granger, 1987]. Both long-run relations are close to unit elastic: $\log \mathrm{PT} = 0.193 + 0.970 \log \mathrm{ES}$ for diesel and $\log \mathrm{PT} = 0.123 + 0.978 \log \mathrm{ES}$ for gasoline.

The adjustment speed, the share of a gap against that relation closed each week, more than quadruples for both products after the transition on the full sample, though the multiple is less firmly pinned than that figure suggests. For diesel it moves from -0.142 to -0.600, shortening the half-life of a deviation from 4.5 to 0.8 weeks. For gasoline it moves from -0.091 to -0.368, a half-life of 7.3 weeks falling to 1.5. Both changes are precisely estimated.

The two products differ in the other channel, and the threshold that carries economic content there is one rather than zero. An elasticity below one means a Spanish move is passed through only in part; above one it is overshot. Diesel sits at 0.713 before the transition, 7.2 standard errors below one ($p < 0.001$), and at 1.375 after, 4.4 above it ($p < 0.001$). The relation does not merely tighten: it crosses from under-transmission to overshoot. Gasoline starts below one at 0.789 ($p = 0.003$) and ends at 1.074, which is not distinguishable from one ($p = 0.742$); its interaction of 0.285 carries $p = 0.227$, so the change itself is not established either.

Linkage to Spain therefore tightened for both fuels, and for both it tightened in the long-run adjustment channel, which a difference-only model cannot observe by construction. For diesel the contemporaneous channel tightened as well. Reporting only the short-run coefficient would have supported the conclusion that gasoline was unaffected, and that conclusion would have been an artefact of the specification.

Table 11: Error-correction models, two-step Engle–Granger, HAC(8), $n = 993$ weeks. Intervals are ninety-five per cent.

Product	Term	Estimate	Interval	p
diesel	Disequilibrium (lagged)	−0.1415	[−0.194, −0.089]	< 0.001
	× post	−0.4580	[−0.624, −0.292]	< 0.001
	post-period level	−0.5995	[−0.757, −0.442]	< 0.001
	Δ log ES	0.7129	[0.635, 0.791]	< 0.001
	× post	0.6624	[0.477, 0.848]	< 0.001
	post-period level	1.3753	[1.207, 1.544]	< 0.001
	post	−0.0090	[−0.013, −0.005]	< 0.001
gasoline	Disequilibrium (lagged)	−0.0911	[−0.161, −0.022]	0.010
	× post	−0.2767	[−0.445, −0.108]	0.001
	post-period level	−0.3678	[−0.521, −0.215]	< 0.001
	Δ log ES	0.7891	[0.648, 0.930]	< 0.001
	× post	0.2846	[−0.177, 0.746]	0.227
	post-period level	1.0738	[0.634, 1.514]	< 0.001
	post	−0.0029	[−0.006, 0.001]	0.092

What this amounts to is a loss of domestic price insulation. Before the transition a gap between the Portuguese and Spanish pre-tax diesel price took about four and a half weeks to half-close. Afterwards it takes under one. A Portuguese price that departs from the Iberian relation is pulled back roughly four times faster than it used to be, and it departs less in the first place, since the contemporaneous elasticity has risen to 1.375. Portuguese pre-tax diesel prices behave less like the output of a domestic market with its own supply conditions and more like a local expression of an Iberian one.

That reading sits naturally alongside the physical result. A country meeting a larger share of its diesel demand from imports is buying at prices formed in the market those imports come from, so there is less scope for a domestic supply position to hold the local price apart. The two findings are consistent and their timing coincides. The design supports no stronger statement. It is not evidence that the physical change caused the price change. The same tightening would follow from a period in which common cost shocks came to dominate both markets, and 2021–2024 was exactly such a period. Distinguishing the two would need refinery-gate or wholesale prices and an import-origin series, neither of which is observed here.

Gasoline is the useful check on that story. Its contemporaneous elasticity does not move, so whatever tightened the diesel relation did not simply tighten everything. But its adjustment speed does move, by a similar factor, and gasoline is the product whose physical position barely changed. That weakens any account tying the price result directly to the physical one, and it is reported here because it is the observation least favourable to the paper's own framing.

The post-transition period alone identifies every interaction in this model. It is 190 of the 994 weeks and contains the 2022 price episode. Re-estimated without 2022 the diesel post- transition speed is −0.430 and the gasoline speed −0.332, against −0.600 and −0.368 on the full sample; estimated on each half of the period they are −0.643 and −0.479 for diesel, −0.405 and −0.361 for gasoline. The direction survives every cut. The magnitude does not: the multiple of the pre-transition speed runs from 2.96 to 4.36 for diesel and from 3.65 to 4.40 for gasoline. What the design supports is that adjustment became several times faster, not that it became precisely four times faster. A half-life of 0.8 weeks should also be read with its unit in mind: it is shorter than the interval between observations, so it says that a deviation is largely gone by the next weekly reading, not that the data can resolve a duration that short.

Two tables now carry intervals where they previously carried standard errors, and one point is easier to see in that form. The gasoline post-period elasticity of 1.074 has an interval of $[0.634, 1.514]$, which is wide enough to contain both the pre-closure value of 0.789 and the diesel post-period value of 1.375. The gasoline estimate is not evidence of no change so much as an estimate too imprecise to distinguish the two, and the second-break test in this section is what resolves it.

One regime boundary is imposed on the price arm where the physical arm uses three, and whether that matters is testable. Adding a second break at March 2022 and testing its three terms jointly leaves diesel unaffected ($p = 0.223$): the single-break model is adequate for it, and the transition and stress regimes are not distinguishable in the diesel price relation. Gasoline rejects ($p = 0.020$), and the reason is worth setting out, because it qualifies the paragraph above.

Split at both dates, the gasoline contemporaneous elasticity runs 0.789 before the closure, 1.679 during the ten-month transition and 1.051 from March 2022. The middle figure is 3.8 standard errors above one. So gasoline's elasticity does move, sharply, in the window between the closure and the shock, and then returns to about one. The single-break model averages those two post-closure regimes into 1.074 and reports no change, which is arithmetically right and substantively misleading. What the two-break model shows is that gasoline behaved like diesel while the system was adjusting to one refinery, and unlike it once the supply shock had arrived. Diesel, by contrast, goes to 1.478 and stays at 1.375. The claim that survives is narrower than the one the single-break model supports: gasoline's elasticity is not permanently higher, not that it never moved.

Two caveats belong with these estimates. The disequilibrium term is a generated regressor, so the second-stage standard errors are conditional on the first-stage cointegrating estimate. HAC covariance [Newey and West, 1987, Andrews, 1991] mitigates but does not remove this [Pagan, 1984]. And the cointegrating vector is estimated over the whole sample while the adjustment speed is allowed to break at the transition, so the long-run relation is held fixed across a date the rest of the model treats as a break.

That assumption can be tested and has been [Gregory and Hansen, 1996]. Allowing the long-run relation itself to shift in level and slope at an unknown date, and comparing the resulting statistic with a null simulated from independent random walks, both pairs remain cointegrated: the diesel statistic is -7.911 and the gasoline statistic -9.097, against a fifth percentile of -4.883 under no cointegration. The verdict does not depend on holding the vector fixed.

The date the search selects is not the one the model imposes. It falls in 2013 for both products, near the middle of the sample, and not at the May 2021 closure. Break searches favour central dates and this one should not be read as evidence that the price relation shifted at the hydrocracker, but it is a reason not to claim the closure as the date at which the long-run relation changed. What the paper claims is a change in adjustment speed after the closure, which is the second stage and is estimated at an imposed date, not a change in the relation itself.

A system estimator asks the same question without choosing a dependent variable [Johansen, 1988, 1991], and it agrees on the part that matters here. The trace statistic rejects no cointegration for both products at every lag order tried, by a wide margin, and the cointegrating vector it recovers is close to the one the two-step estimator uses: a normalised slope between 0.970 and 0.974 for diesel against the Engle–Granger 0.970, and between 0.958 and 0.967 for gasoline against 0.978. The long-run relation is not an artefact of which series was placed on the left.

It does not agree about everything. The same test also rejects a rank of one, which in a two-variable system means full rank, and full rank means the levels were stationary and no error-correction form was needed. That contradicts the unit-root evidence in Section 9.1, where neither ADF nor KPSS supports stationary levels for either product. The obvious remedy does not work. A small-sample correction to the trace statistic scales it by a factor that depends on the sample length, and at nearly a thousand weekly observations that factor is within one per cent of unity, so every verdict survives it unchanged. The distortion at issue here comes from the frequency and the autocorrelation of the data and not from a short sample, which is why correcting for sample length leaves it in place. The univariate evidence is given precedence for that reason. The disagreement is recorded because it is the kind of thing a reader should be told about.

9.5 Placebo tests

Two objections to the price result can be tested instead of argued about, and the bulletin supplies the material for both because it prices every member state on the same definition.

The first is that nothing Portuguese need be involved. If the disequilibrium term simply became more volatile after 2021, every European pair priced against Spain would show a faster adjustment speed at that date. Running the identical specification on four neighbours that closed no refinery gives ratios of 0.99 for France ($p = 0.990$), 1.11 for Belgium ($p = 0.624$), 0.15 for Germany ($p < 0.001$, in the opposite direction) and 4.17 for Italy ($p = 0.064$), against 4.24 for Portugal ($p < 0.001$). Three of the four show nothing or the reverse. Italy shows a comparable ratio on much weaker evidence, and that is a real qualification, not one to be explained away: whatever moved the Portuguese relation is not unique to Portugal, though it is far from general.

The second is that the date is arbitrary. A false break tested on the full sample is not a test, because everything after any pre-2021 date includes the closure and recovers a diluted version of the real effect. Estimated on the pre-closure sample only, false breaks at May 2011, 2013, 2015, 2017 and 2019 give ratios of 0.25, 0.34, 0.60, 1.12 and 1.60. The two significant ones run the wrong way, the adjustment speed falling where the real break raises it. The sequence is also monotone in the date: the closer a false break sits to May 2021, the larger its ratio. That is what a ramp looks like, and it means the honest statement is narrower than a flat denial. No date before May 2021 produces anything close to 4.24, and the two nearest to it are the weakest placebos, at 1.12 and 1.60 with $p = 0.674$ and $p = 0.098$. If the tightening began before the closure and accelerated at it, this design cannot separate the two.

9.6 Price spreads

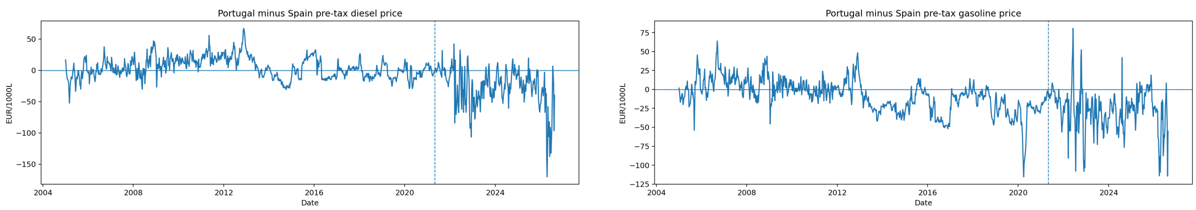

Figure 9: Portugal and Spain weekly pre-tax prices and their spread: diesel (left), gasoline (right).

The mean Portugal-minus-Spain pre-tax diesel spread moves from +4.12 to −26.71 EUR per 1000 litres across the transition, a change of −30.83. The gasoline spread moves by −18.90. These are descriptive.

The pooled unit-root test on these spreads is the wrong one to quote. The spread mean shifts across the transition, and an unmodelled level shift biases a unit-root test towards not rejecting [Perron, 1989], so a pooled verdict understates mean reversion by construction. Treating the break date as unknown, not imposed, is the other standard response [Zivot and Andrews, 1992]; here the date is documented, so the regimes are split at it. Run within regimes, both spreads reject in both periods: diesel at $p = 0.0025$ before and below 0.001 after, gasoline at $p = 0.0104$ and below 0.001. The pooled diesel figure of $p = 0.043$ is the attenuated version of that, and reads as a weaker finding only because the break is unmodelled. Each spread fluctuates around a level within a regime, and the level itself moves between them.

None of this is a second opinion on Section 9.1. The raw difference PT − ES is the cointegrating combination only if the long-run coefficient is one and the relation is estimated in levels rather than logs. The second condition fails for both products. The first fails for diesel, where dynamic least squares puts the slope between 0.969 and 0.972 and rejects one at $p \leq 0.005$ across every lead-and-lag span tried. It does not clearly fail for gasoline, where the same estimator gives 0.970 to 0.977 and rejects one only at the longest span ($p = 0.026$ at eight leads and lags, against $p = 0.097$ at two). For gasoline the EUR spread and the modelled residual are closer to the same object than they are for diesel, and the separation between them rests on the logs-against-levels difference alone. These spread tests remain descriptive companions to the figures above.

10 Portugal–Spain Comparison

10.1 The physical divergence is Portugal-specific

Spain refines the same products and was exposed to the same European energy shock in 2022, so its diesel balance is a useful, if imperfect, reference for what a shock common to the Iberian market did to a neighbouring system. Table 12 sets the two side by side.

The reason to look at Spain is specific. The two countries share a long land border, a product market and a regulatory environment, and fuel moves across that border in response to price and tax differences [Leal et al., 2009, Banfi et al., 2005]. They faced the same withdrawal from Russian products on the same timetable, and Spanish prices are already the comparison series in the price models, so the same country carries both halves of the paper. What differs is the event. Portugal closed a refinery in May 2021 and Spain did not. If the post-2021 Portuguese movement were the European shock working through any Iberian refining system, Spain should show something like it. Spain shows nothing like it. The comparison establishes that much and no more.

Table 12: Diesel balance ratios, Portugal against Spain.

	Refinery output / demand			Net imports / demand		
Year	PT	ES	PT−ES	PT	ES	PT−ES
2018	1.049	0.899	+0.150	−0.056	−0.046	−0.010
2019	0.905	0.902	+0.003	+0.070	−0.032	+0.102
2020	1.077	0.932	+0.145	−0.084	−0.051	−0.033
2021	0.912	0.847	+0.065	+0.026	+0.009	+0.016
2022	0.867	0.904	−0.037	+0.140	+0.001	+0.139
2023	0.678	0.923	−0.245	+0.262	−0.041	+0.303
2024	0.855	0.898	−0.043	+0.105	−0.040	+0.145

Spanish diesel output covers between 0.85 and 0.93 of Spanish demand in every year shown, with no trend over those years: Spain absorbed 2022 without a change in its physical balance. The window is 2018–2024 because that is the span over which the comparison is being made, not because Spain looks flat in it. Spain's ratio is not flat across the full panel, falling to 0.667 in 2007 before recovering, and Figure 10 shows the whole series so a reader can see how much of

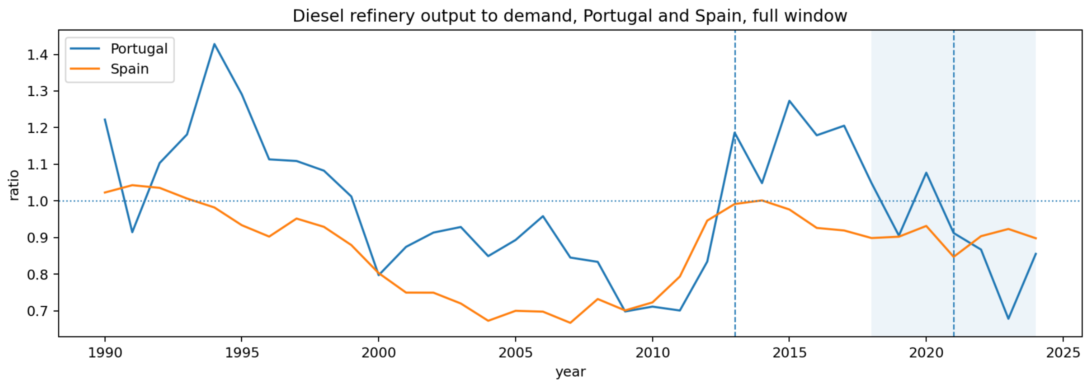

Figure 10: Diesel refinery output to demand for Portugal and Spain over the full window. The shaded band is the 2018–2024 span of Table 12; the dashed lines mark 2013 and 2021. Spanish coverage is not flat outside the shaded years.

the apparent stability is the choice of window. Portugal covered more of its own diesel demand than Spain did in 2018, 2020 and 2021, and less from 2022, falling to 0.678 in 2023 while Spain sat at 0.923. Across those years Spanish net imports stay within 0.052 of balance. Portugal reaches 0.262 in 2023.

That 0.052 has to be set against Section 10.2, and it does not survive the comparison. The Spanish diesel balance residual runs from −14.7 to +13.9 per cent of demand, so the accounting identity closes to within roughly 0.15 of demand at worst, three times the band being called stable. A movement of 0.052 in Spanish net imports is therefore not measurable in this data, and no weight should be placed on Spanish flatness as a quantity. What survives the residual is the Portuguese movement: the coverage ratio falls by about 0.40 between 2020 and 2023, against Portuguese residuals reaching −9.1 and +12.0 per cent. The contrast between the two countries is a contrast between a movement larger than the measurement error and one smaller than it. This is weaker than a contrast between movement and stability, and it is what the data supports.

This does not identify a causal effect, and Spain is not used as a control. A control would have to be a system alike in every relevant respect except the treatment, and two national refining sectors never are. It weakens the common-shock explanation without disposing of it, because a divergence appearing in one country and not the other is not by itself sufficient evidence that a shared shock was unimportant, because the two systems differ in ways this paper has not measured. Crude slate, refinery maintenance schedules, inventory positions, demand composition and product-market exposure all differ, and the Russian VGO disruption in particular did not fall on the two countries equally. Establishing that the common shock was second-order would require diagnostics on each of those margins, none of which are estimated here.

Three alternatives remain live, and they are not equally troubling. The first is that Sines was more exposed to the specific feedstock that was withdrawn than the Spanish system was, in which case the divergence is about feedstock sourcing and not about capacity at all. The second is maintenance, where an unplanned outage at the one remaining Portuguese site would produce a similar signature and is not observable in annual balances. The third is demand composition, since the two countries recovered from the pandemic on slightly different paths, and their fuel demand elasticities are estimated separately [Bakhat et al., 2017].

The first two are worth separating from the common-shock story proper. Both run through the fact that Portugal had one conversion site by 2022, and having one site is a consequence of the closure. An account in which a common European shock hit Portugal harder because its refining had just been concentrated is not an alternative to the closure explanation. It is a version of it, with the closure acting on the propagation of the shock rather than on the level of output directly. That folding depends on Matosinhos having been able to offer a material offset, and Section 2.1 records that the two sites' configurations are not observed here and that how much flexibility was lost is unknown. It is a defensible reading of the mechanism, not something the data entails. Only the third alternative, demand composition, is genuinely independent of the closure, and the 2022 diesel demand figure sits close to its own baseline, giving it weak support.

10.2 Balance residuals

The Eurostat panel permits an internal consistency check absent from JODI. Defining the residual as $Q^{\text{refinery}} + M - X - D$, the residuals are large and do not sit on one side of zero. Across 1990–2024 the Spanish gasoline residual ranges from -20.0 to $+15.2$ per cent of demand and the Spanish diesel residual from -14.7 to $+13.9$. The Portuguese ones are smaller but still reach -9.1 and $+12.0$. They reflect stock changes, refinery fuel and statistical differences that the four reported terms do not close. Balance residuals are therefore reported but not interpreted as flows, and no claim in this paper is derived from a residual. Their size is also why the paper infers nothing about stock behaviour from the balance, since a residual of this magnitude would swamp any inventory signal it might carry.

11 Robustness

11.1 Event windows

The pre/post comparisons use five pre-event and three post-event years. Repeating them over three window pairs leaves every conclusion unchanged.

Table 13: Pre/post difference around 2021 by window, 2021 excluded.

Product	Outcome	3 pre / 2 post	5 pre / 3 post	8 pre / 3 post
diesel	net imports / demand	+0.224	+0.277	+0.323
diesel	refinery output / demand	−0.238	−0.283	−0.315
gasoline	net imports / demand	+0.333	+0.393	+0.281
gasoline	refinery output / demand	−0.388	−0.424	−0.321

Every estimate keeps its sign across all three windows. The diesel ratios grow monotonically with a longer pre-window. This is expected, because a longer baseline reaches further into the pre-hydrocracker regime and widens the measured gap. The gasoline estimates move the other way, shrinking as the pre-window lengthens, because the gasoline surplus was at its widest in the middle of the window, not immediately before the transition. No conclusion here depends on the window choice.

On the headline five-and-three window, diesel gross import dependence rises from 0.202 to 0.278, a change of 7.6 percentage points. The diesel net-import-to-demand ratio rises from -0.108 to 0.169, a change of 27.7 percentage points, and the diesel refinery-output-to-demand ratio falls from 1.083 to 0.800. These are descriptive pre/post differences, not effect estimates.

11.2 Two further robustness questions

Two checks that would otherwise interrupt the argument are reported in Appendix B. The first re-estimates every 2022 result on each of the two trade sources, since the disputed cells fall in that year. One result changes significance and it is marked wherever it appears. The second applies a Benjamini–Hochberg correction across all twenty-four Chow tests, and six survive it.

12 Limitations

1. **2022 confounds the post-closure window.** The Matosinhos transition and the European energy shock fall ten months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance and the Russian VGO disruption are not separately identified. Two of those run through the concentration the closure produced, and on the reading set out in Section 13 they are versions of it. That reading assumes Matosinhos would have offered a material offset, which Section 2.1 records as unobserved, so it is a defensible interpretation of the mechanism. Unplanned maintenance is the one that is genuinely independent and the one no series here can rule out.
2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result is an association measured around a documented event. The Spain comparison constrains the common-shock explanation but does not license difference-in-differences: no pre- trend or common-shock diagnostics have been estimated.
3. **DGEG trade covers only 2019 –2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission and is not independent of it.
4. **Low annual power after 2021.** Three post- event annual observations cannot support a Chow test. The interrupted-trend models carry the annual evidence for 2022, and they impose a common trend that the Chow design does not.
5. **The two arms disagree about refinery output.** The annual balance reads Eurostat and the monthly models read JODI, and aggregated to years the two differ on diesel refinery output by up to 9.1%, above the annual source through the 2010s and below it from 2020. Refinery output carries the monthly phase estimates and the headline coverage ratio, so the magnitude of both should be read with that difference in mind. See Section B.2.
6. **The post-transition period carries the whole price result.** Every interaction in the error- correction models is identified off 190 weeks, a fifth of the sample, and those weeks contain the 2022 price episode. Re-estimating without 2022 and on each half of the period leaves the direction intact and moves the multiple of the pre-transition adjustment speed between 2.96 and 4.40. A longer post period is the only thing that would settle it.
7. **Generated regressor in both error-correction models.** The disequilibrium term is estimated in a first stage, so the reported second-stage standard errors are conditional on it. A single-step or maximum-likelihood estimator would price that uncertainty properly. The related worry, that one cointegrating vector is assumed to hold across the transition, is tested in Section 9.4 and survives: both pairs remain cointegrated when the relation is allowed to shift at an unknown date.
8. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose exact semantics have not been confirmed against JODI documentation.
9. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. The price results describe retail co-movement, not refinery-gate or international wholesale transmission, and are not pass-through in the causal sense.
10. **Balance residuals are not closed.** The four Eurostat terms do not sum to zero and stock changes are not modelled. The Spanish diesel residual reaches 15 per cent of demand, which is larger than the Spanish movements the comparison in Section 10 rests on, so that comparison is qualitative. The Portuguese movement is several times its own residual and is not affected the same way.

13 Conclusions

13.1 What changed in Portugal's dependence profile

Over thirty-five years Portugal's diesel system crossed the same line three times and ended further from self-coverage than it began. It was a net exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013 after the hydrocracker in every year but 2019, and an importer in every year since 2021. By 2023 domestic manufacturing covered 0.678 of diesel demand and the net-import-to-demand ratio stood at 0.262. Those are the highest readings in the window, but only just: 2009 reaches 0.2534 and 0.6982, and the gap between the two years is a fraction of the 0.1487 standard deviation of the annual change in that ratio. The claim worth making is not that 2023 set a record. It is that 2023 ran the dependence of 2009–2011 with one refinery instead of two, which the margins support.

Read against the four senses of resilience set out in Section 2, the profile changed on all four and in the same direction. The domestic buffer thinned, in that the diesel refinery-output-to-demand ratio fell below its 1990s level. Import exposure rose on both measures, in volume and in net position. Manufacturing concentrated from two sites to one. Price integration tightened, on both the contemporaneous and the adjustment channel. Nothing moved the other way.

Gasoline crossed none of them, which is a different statement from saying it did not move. On the annual ratios it moves at least as far as diesel: across the 2021 closure its net-import-to-demand ratio shifts by +0.393 against diesel's +0.277 and its refinery-output-to-demand ratio by −0.424 against diesel's −0.283, in the five-and-three window and in the same direction in every other. Its price adjustment speed quadruples much as diesel's does. What separates the two is where they started. Gasoline entered every one of these events manufacturing roughly twice its domestic demand and left still manufacturing well above it, so a movement of that size crossed no threshold, while diesel entered with a ratio near one and the same proportional loss carried it across. That is the practical lesson available here: what determined exposure was not the size of the movement but how much margin the product had before it began.

The same diesel-gasoline asymmetry appears in this country pair under an entirely different shock. A Portuguese fuel tax rise raised diesel sales in Spanish border provinces by six to nine per cent with no comparable effect on gasoline, attributed to heavy goods vehicles with large tanks [Teixidó et al., 2024]. That the asymmetry recurs under a tax shock as well as a capacity shock supports reading it as a property of how the two products are used, not of either event. It also names something the price models here do not control for: a live, diesel-specific cross-border channel between exactly the two markets whose prices are being compared.

13.2 The best explanation available

Putting the pieces together, the account this paper can defend is one of interacting causes, and the interaction is what makes the period distinctive.

Portugal's diesel position was never a fixed property of the country. It was the running balance between a demand structure that had dieselised faster than the refining system could follow and a refining configuration that was periodically adjusted to catch up. The 2013 hydrocracker was such an adjustment, and it worked: it moved the country back into diesel surplus for most of the following decade. The 2021 closure was an adjustment in the other direction, and it did more than remove capacity. It concentrated what remained in a single site, so that both the volume of national diesel manufacturing and its feedstock exposure came to rest in the same place.

Ten months later a shock arrived that was specific to that feedstock. In the earlier configuration it would have hit one of two plants. In the new one it hit all of Portuguese refining, and the balance shows the consequence exactly: diesel manufacturing at the bottom of its baseline range, gasoline manufacturing inside its own, demand close to baseline, exports at the zeroth percentile and imports at the hundredth. The system met the shock, but it met it entirely through the trade channel, because after May 2021 there was no other channel to meet it through.

The import series is worth reading carefully here, because it does not single 2022 out. Diesel imports were higher in 2021, the closure year, than in 2022, and higher again in 2023 than in either. What 2022 contributes is not the largest import volume in the window but the clearest picture of the mechanism, because it is the year in which the four terms of the balance move against each other in a way that identifies where the constraint bound. On that reading the closure and the shock are not competing explanations for 2022, and it is a mistake to ask which mattered more. The closure determined how a subsequent shock would propagate. The shock determined when that would be tested. Gasoline is the nearest thing to a control the data supplies: it went through the same closure at the same plant and shows almost none of it in the physical balance, because it entered with a margin that diesel did not have. It is not a clean control, because its price adjustment speed moved by much the same factor as diesel's, and that is the observation least favourable to the account given here.

13.3 What can and cannot be inferred

The causal question is open and this design cannot close it. The monthly model finds both phases independently associated with lower refinery output and higher net import dependence, with the 2022 shift the larger of the two. The evidence goes no further. It is worth being clear that no better estimator would close the gap either, for the reason given above: the decomposition being asked for is not a property the physical system has.

Nor can the paper say how much of the post-2021 level is permanent. The monthly model bears on this more than the annual series do. The post-stress phase sits -110.52 kt per month below the pre-closure counterfactual in refinery output and $+0.3174$ above it on the ratio, both firmly estimated, so the displacement is still there in 2023 and 2024. But both phase trends point back towards the counterfactual, at $+3.67$ kt per month ($p = 0.004$) and -0.0086 ($p = 0.020$). The level is displaced and closing, which is more than the annual series can say and less than an answer. Coverage recovers from 0.678 in 2023 to 0.855 in 2024, which is consistent with 2023 having been the trough of an adjustment and equally consistent with a system that now runs closer to its import channel and fluctuates around it.

What follows for policy is correspondingly narrow, and narrower than the subject usually invites. The structural exposure measures moved, and they moved together and in one direction. A policy discussion would need to start from that. But this paper has not observed stocks, storage, port capacity, supplier diversity or the terms on which the additional imports were obtained, and those are the quantities that determine whether higher import exposure is a vulnerability or simply a different and possibly cheaper way of meeting demand. A country importing a large share of its diesel through deep and well-connected markets may be better placed than one manufacturing more of its own from a single exposed feedstock route. Nothing here can tell those two states apart, and a reader who takes rising import dependence as *prima facie* evidence of reduced security is going beyond what is shown.

The price finding carries the same limit in a different form. Portuguese pre-tax diesel prices now track Spanish ones more closely and revert to the Iberian relation several times faster. That is co-movement between two retail series, not pass-through, and the specification controls for nothing the two markets have in common. Nothing here speaks to refinery-gate or wholesale prices, which are not observed at all.

13.4 What evidence would settle it

The obvious constraint is time. Three post-closure annual observations cannot support a Chow test, and several results in this paper are weak precisely because the post-event window is short. A few more years would let the annual designs speak, and would answer the question left open above about whether 2023 was a trough or a level.

Three other kinds of evidence would do more than waiting. A monthly series of Sines throughput and unplanned outages would separate a feedstock disruption from a maintenance event. Annual balances cannot tell those apart, and maintenance is the most troubling of the remaining alternatives to the account here. An import-origin series would show whether the additional diesel came from the same markets that set the Spanish price. The price result is consistent with that mechanism without demonstrating it. And refinery-gate or wholesale quotations would let the price question be asked as pass-through rather than as co-movement, the form in which it is worth asking.

Failing all of those, the cleanest natural experiment would be a comparable single-refinery closure elsewhere in Europe at a date not adjacent to a continent-wide supply shock. The identification problem in this paper is not a shortage of data about Portugal. It is that the two events fall ten months apart, and no amount of Portuguese data will move them.

Claim–Evidence Matrix

Table 14: Every claim the paper makes, where it is shown, the strength it is asserted at, and what remains against it.

Claim	Where shown	Level	Remaining caveat
Sines hydrocracker entered commercial production in 2013	Section 1	documented event	Corporate disclosure, not outcome data
Matosinhos refining ceased May 2021	Section 1	documented event	Announcement date differs from closure date
2022 Russian product and VGO disruption	Section 1	documented event	Establishes timing, not magnitude
Diesel demand averaged 4,522 kt, output 4,348 kt	Section 5.1	descriptive	Window means, not a model
Portugal was a net diesel exporter in every year 2013–2018, and in 2020	Tables 2, 3	descriptive	Ratio is not self-sufficiency; 2019 is a net-import year
Gasoline is a net export product in every year but 1993	Tables 2, 3	descriptive	Same caveat; 1993 net-import ratio +0.056
Balance at each event year	Table 4	descriptive	2021 marked transition
Refining regime by year	Table 4	documented	Regime label, not capacity
Diesel net-import ratio breaks at 2013	Table 5	structural break	Diagnostic, not a causal estimator
No Chow break detected at 2022	Table 5	structural break	Three post-event years; low power
2000 break is exploratory, not pre-specified	Table 5	association	Added after extending the panel; no conclusion rests on it
Interrupted trends do detect 2022	Table 6	association	Imposes a common trend; 2022 export rows depend on disputed cells

Table 14: Every claim the paper makes, where it is shown, the strength it is asserted at, and what remains against it.

Claim	**Where shown**	**Level**	**Remaining caveat**
Diesel 2022 annual results do not depend on the disputed cell	Table 16	association	Annual panel uses the corroborated source throughout
No 2022 gasoline annual result reaches significance	Table 16	—	Weak on either source; not relied on
Monthly phase means shift after May 2021	Table 7	descriptive	No seasonality control
Refinery output sits −70.52 kt/month below the pre-closure counterfactual in the transition	Table 8	association	Not distinguishable from zero ($p = 0.084$); quote with its phase-trend term
In the stress phase it sits −101.10 kt/month below the same counterfactual	Table 8	association	Not incremental to the transition shift; confounded with demand recovery
Monthly models cover 276 months, 2002–2024	Section 3.2	descriptive	JODI begins in 2002, later than the annual window
2022 diesel exports at the 0th baseline percentile	Table 9	descriptive	Sources disagree on this cell; both reported
Diesel log prices are non-stationary; gasoline rejects only marginally	Section 9.1	descriptive	ADF only; the gasoline verdict changes with the specification
Both pairs are cointegrated in logs, diesel $p < 0.001$ and gasoline $p = 0.0026$	Section 9.1	descriptive	Engle–Granger; survives allowing the relation to shift at an unknown date
EUR-levels model gives the opposite sign and is not used	Section 9.2	—	Retained as a specification warning only
Diesel contemporaneous elasticity rises 0.7129 to 1.3753	Tables 11, 10	association	Retail, not wholesale, prices; 1.00 in the difference-only model
Gasoline contemporaneous elasticity not permanently higher	Table 11	association	Interaction $p = 0.227$ on one break; it does move in the transition window, see Section 9.4
Both adjustment speeds increase several-fold	Table 11	association	Quadruple on the full sample; the multiple falls on every post-period subset, see Section 9.4
2022 diesel export level shift of −781.7 kt	Table 6	association	Falls to −570.0 and $p = 0.078$ once the 2013 break is held constant
2000 break in diesel imports	Section 6.1	descriptive	Exploratory; sup-Wald over searched break dates gives $p = 0.088$
PT–ES diesel spread moves −30.8 EUR/1000L	Section 9.6	descriptive	Descriptive; the spread is not the modelled residual
Both spreads mean-revert within each regime	Section 9.6	descriptive	Pooled ADF is attenuated by the level shift; within-regime $p \leq 0.011$

Table 14: Every claim the paper makes, where it is shown, the strength it is asserted at, and what remains against it.

Claim	**Where shown**	**Level**	**Remaining caveat**
Portugal's diesel balance falls; Spain's shows no comparable move	Table 12	descriptive	Spanish flatness is smaller than the Spanish balance residual, so only the Portuguese side is measurable
Eurostat balance residuals are not closed	Section 10.2	descriptive	Stock changes not modelled
Pre/post differences hold across three windows	Table 13	descriptive	Magnitudes vary with baseline length
Headline pre/post dependence change	Section 11.1	descriptive	Not an effect estimate
JODI coverage is complete for every year	Table 1	descriptive	Assessment-code semantics unverified
Sources agree on 89% of full-window trade cells	Section B.1	descriptive	Eurostat is DGEG-derived
Sources and retrieval points are frozen	Section 3.1	documented	Snapshot of one vintage
Hypotheses were pre-specified	Section 1	documented	Recorded before estimation
No causal effect of the closure is claimed	—	—	2022 is not separable by this design

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A Data and Code Availability

All data underlying this paper are public and are listed in the References. The analysis code, the derived tables behind every figure and table above, and a document recording the provenance of each one are available in the project repository [Ribeiro, 2026].

Results depend on the software that produced them as much as on the inputs, and a checksum over outputs records neither. The archive therefore carries the environment: Python 3.13.5 with NumPy 2.3.5, pandas 2.3.3, SciPy 1.15.3 and statsmodels 0.14.5. Every model in this paper is ordinary least squares from `statsmodels`, with heteroskedasticity- and autocorrelation-consistent covariance at one lag for the annual models, three for the monthly ones and eight for the weekly ones. Unit-root tests are the augmented Dickey–Fuller and KPSS implementations from the same library, with lag length by AIC unless a sensitivity states otherwise; cointegration is Engle– Granger from `statsmodels`, and the long-run slope test is dynamic least squares with leads and lags of the differenced regressor. The break-date search simulates its own null, so it needs no published table to be reproduced.

Derived results are archived with a checksum manifest that accounts for every file in the archive, so the tables and figures reproduced here can be verified against the exact inputs used to produce them. Eight automated readiness checks are applied before any figure is reported: they verify archive completeness, annual coverage of the monthly source, validity of the monthly panel and its models, availability of the cross-country balance, completeness of the price diagnostics including the recorded model-family verdict, and source reconciliation. The reconciliation check additionally requires every out-of-tolerance cell to be named with a reason, so a later data vintage cannot widen the gap between sources silently. All eight pass for this paper.

B Source Reconciliation, Sensitivity and Multiplicity

This appendix holds the material that establishes the evidence is sound: how the sources are compared, what happens to the results when a disputed cell is replaced, and how the multiple testing is corrected. It is placed here because it answers a question about the evidence, where the body answers questions about Portugal, and every claim it supports is stated in the body with a pointer back to this appendix.

B.1 Source reconciliation

JODI and DGEG are the only genuinely independent pair. Across the 24 overlapping 2019–2024 trade cells, 66.7% agree within tolerance. Over the full window, JODI and Eurostat agree on 89% of 80 cells, with only three of the nine divergences occurring before 2019.

A cell is flagged only when it breaches both the absolute and the percentage tolerance. Applying them as alternatives makes a 25 kt floor bind on series of roughly 1,600 kt, an effective tolerance of 1.6%, which would flag agreement as close as 2.1% as failure. Each of the eight remaining divergent cells is enumerated in the analysis configuration, together with the series Eurostat identifies as the outlier: six where JODI is out of line, two where the DGEG workbook is. An unlisted divergence fails the readiness gate however few cells it touches.

B.2 The two arms against each other

The reconciliation above compares trade cells between sources. It leaves the two arms of this study uncompared, and they measure the same physical quantities: the annual balance reads Eurostat, the monthly event models read JODI. The readiness gate checks that the monthly source covers the annual window. Coverage establishes that months are present and says nothing about whether the two sources agree on what is in them. Table 15 aggregates the monthly panel to complete calendar years and sets it against the annual panel on all four balance terms.

Table 15: Monthly source aggregated to complete years against the annual panel, both products, 2002–2024. Agreement is within five per cent.

Flow	Cells	Median |diff| %	Max |diff| %	Within tolerance
Exports	46	0.16	46.80	0.913
Imports	46	1.08	40.00	0.804
Refinery output	46	2.35	9.09	0.826
Demand	46	1.11	3.07	1.000

Demand agrees everywhere. The two largest trade disagreements are cells already known and already carried through a sensitivity: 2022 diesel exports at −46.8%, which is the disputed cell of Section B.3, and 2002 gasoline imports at −40.0% on a base of 60 kt.

Refinery output is the one that matters and the one that was never checked. It never reaches the size of the disputed trade cells, but it does not behave like noise either. The monthly source runs above the annual source through the 2010s, by 9.1% at its widest in 2010, and below it from 2020 onward, by 8.0% in 2022. The sign changes at the boundary of the period this paper is about. Which source is right is not established here; what matters is that a level comparison spanning that boundary inherits a source difference of roughly eight percentage points, and that the monthly phase estimates of Section 7 are level comparisons spanning exactly it. They are fitted entirely within JODI, so they are internally consistent, but a reader should treat their magnitude as carrying that uncertainty on top of the standard errors reported.

B.3 Sensitivity to the disputed trade cells

Two 2022 trade cells are disputed between the two independent sources: diesel exports, where the monthly source reports 331.0 kt against 622.2 kt nationally, and gasoline exports, where it reports 1,346.0 kt against 1,501.7 kt.

Moving the annual panel to Eurostat changes what that disagreement can affect. The annual results are now computed from the national submission, which already carries the corroborated value for the diesel cell, so the disputed monthly figure does not enter the annual arm at all. Table 16 records this: the diesel rows are identical under both sources because there is nothing to swap. The disagreement now bears only on the monthly arm and on the JODI cross-check, not on the annual conclusions.

The diesel results are unaffected by the swap, for the reason just given, and both remain significant at conventional levels.

The gasoline results are weak on either source. Neither the export level change nor the net- import-ratio shift reaches five per cent under the primary values, and swapping to the corroborated ones reduces both further. No 2022 gasoline result is relied on anywhere in this paper. The gasoline evidence for that year rests on the monthly design of Section 7.

Table 16: 2022 interrupted-trend level changes, computed on each trade source.

		Primary source		Corroborated	
Product	Outcome	Estimate	p	Estimate	p
diesel	exports	−781.7	0.022	−781.7	0.022
diesel	net imp./dem.	+0.201	0.031	+0.201	0.031
gasoline	exports	−270.9	0.121	−202.8	0.149
gasoline	net imp./dem.	+0.307	0.095	+0.253	0.111

B.4 Multiplicity

Twenty-four Chow tests are run and Benjamini–Hochberg adjustment is applied across all of them, not to a selected subset. Six survive at conventional levels. That procedure assumes the tests are independent or positively dependent, and these are neither, since the net-import-to-demand ratio is a deterministic function of the imports and exports tested beside it. Benjamini–Yekutieli [Benjamini and Yekutieli, 2001] holds under arbitrary dependence at the cost of a factor of 3.776 here, and four survive it. The two that do not are the 2013 diesel refinery-output break, at 0.015 under Benjamini–Hochberg and 0.057 under Benjamini–Yekutieli, and the 2000 diesel import break, at 0.043 and 0.163. The two results the paper leans on, the 2013 diesel export and net-import breaks, survive both at 0.0007. The monthly models are not included in that adjustment and their p-values should be read as unadjusted; the phase terms that carry the conclusions, on refinery output and on the net-import-to-demand ratio, have $p \leq 0.007$, inside any reasonable correction for the twelve terms reported. The import-level terms are weaker and are not relied on: the Matosinhos import shift is significant only at the ten per cent level.